Evaluation of Immune Management Strategies to Control and Eliminate Porcine Reproductive and Respiratory Syndrome Virus (PRRSv)

A THESIS

SUBMITTED TO THE FACULTY OF

UNIVERSITY OF MINNESOTA

BY

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IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR THE DEGREE OF

DOCTOR OF PHILOSOPHY

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February, 2013

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Acknowledgements

Coming from Brazil to Minnesota was a decision that my wife and I made together. I really appreciate all the support that she gave to the kids and me during this incredible journey in the land of 10,000 lakes. Leticia was a wonderful wife, 2x mom and best friend. Without her support, care and love the PhD journey would never have started. Thank you for believing we could make it. I am the happiest husband and dad because of you Leticia, Pedro and Davi!! Thank you for all the motivation – you are the best family ever.

PhD is the highest scientific degree that one can achieve. I am really excited to earn my PhD at the prominent University of Minnesota's swine group. However, more important than the scientific degree are the valuable relationships that I have developed along the way. I am really proud of the friendships my family and I developed during our time in Minnesota. Therefore, I would like to acknowledge the individuals that contributed significantly to my success on the graduate program:

First of all, thank you very much Dr. Montse for advising me throughout my PhD journey and for turning me into a scientist. You not only cared about my work, but also about my family and me. It was my pleasure to count with you and with your experience along the way!

I think I never left Dr. Bob's office without learning something new or fun. Thank you for your advice, leadership and inspiration Bob. I am glad I had the opportunity to work with you.

Dr. Scott, thanks for accepting me for the PhD position, for the guidance during the ATP trial and for all the experience that you shared with me during our discussions and trips.

Drs. Albert and Murtaugh were great scientists who inspired me, allowing a better understanding of the worlds of diagnostics, immunology and molecular biology. Thank you for accepting to join my PhD committee.

Dr. Jean Paul Cano, I appreciate all the time and patience you had with me during all the studies described in this thesis. I learned a lot from your knowledge and leadership.

Many thanks to Drs. Peter Davies, Tom Molitor, Han Soo Joo, Randy Singer, John Deen, Marie Gramer and William Thomas for taking time for insightful discussions and for sharing their knowledge with me and fellow graduate students.

This thesis would not have been possible without the collaboration of veterinarians and producers. I really appreciate their time, patience and help.

To all friends - I really appreciate the moments we shared together in Minnesota talking about pretty much everything from science to the art of raising kids, from beers to statistics, from past experiences to future plans! Special thanks to the folks at school Corzo, Fabio & family, Anna, Seth, Matt, Nubia, Maria, Steve, Carmen, Andres, Niti and Victor. Also many thanks to the great CTC neighbors Luis & family, Joao & family, Carles & family, Diego & family and to our great friend Marina.

Dedication

This thesis is dedicated to my beloved parents, Lucia & Guido Linhares. Thank you both for all the education, love, examples and care that you provided to me, Du and Bruna. I am really proud of being your son and I dedicate this work to you.

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General introduction

Porcine reproductive and respiratory syndrome (PRRS) is an infectious disease caused by PRRS virus (PRRSv). PRRSv is considered one of the most economically important infectious agents in the swine industry worldwide (2011; Neumann et al., 2005; Nieuwenhuis et al., 2012) and it has been detected in all pig producing countries with a few exceptions (Zimmerman et al., 2006). Unless eliminated, PRRSv infection can persist for prolonged periods of time in breeding herds with continuous replacement pig introductions due to presence of pigs at various stages of infection and immunity (Dee et al., 1996; Dee et al., 1997a).

Because PRRSv is difficult to control, the US swine industry is headed towards PRRSv elimination from well-defined geographical regions (Corzo et al., 2010). Success of PRRSv regional elimination programs depends directly on the ability to clear infection from individual breeding herds. Thereby, strategies to control and eliminate PRRSv from production systems have been documented, including whole herd depopulation-repopulation (Dee, 2003), test-and-removal (Dee and Molitor, 1998), segregated early weaning (Rajic et al., 2001) and herd closure (Torremorell et al., 2003). The later method has become a common practice in the U.S. swine industry (Zimmerman et al., 2012).

As part of the herd closure program some veterinarians advocate deliberate exposure of all breeding pigs in the herd with PRRSv in an effort to increase herd immunity (Desrosiers and Boutin, 2002; Voglmayr et al., 2006). In conjunction with deliberate exposure, the program includes filling the facility with extra replacement pigs and then temporarily closing the breeding herd to any replacement pig introduction.

Hence, the herd closure program is commonly thought of as having “load, close and expose” components.

There are different immunogens that can be used for whole-herd exposure including the resident live virus and attenuated virus vaccine. To our knowledge, there is no scientific data comparing the effectiveness of different immunization protocols to reach PRRSv-stability on PRRSv-positive herds under field conditions. Likewise, it has been suggested that enforcement of specific management practices reduces within-herd transmission of pathogens (McCaw, 2000). However, it is unknown whether management practices adopted in breeding herds during herd closure or other risk factors are associated with success of PRRSv elimination programs.

Studies are needed to compare immunization methods as part of herd closure programs in terms of time required to reach PRRSv stability and success rate of PRRSv elimination using whole herd exposure methods combined with herd closure. Moreover, there is no comparison of whole herd exposure methods in terms of production impact or overall economic advantage following intervention.

Success of regional PRRSv elimination programs also depends on the ability to minimize virus dissemination from infected pig populations. Christopher-Hennings et al reported that boars previously vaccinated had reduced duration of wild-type PRRSv shedding via semen (Christopher-Hennings et al., 1997). Additionally, it has been reported that attenuated PRRSv vaccine decreases duration of viremia (Zuckermann et al., 2007), decreases the severity of PRRS clinical signs (Martelli et al., 2009) and shortens PRRSv shedding (Cano et al., 2007a) in pigs that were subsequently infected with virulent (“wild-type”) PRRSv. However, little is known

about the use of modified-live virus (MLV) vaccines on viral shedding of pig populations that are acutely infected with PRRSv.

The goal of this thesis is to evaluate immune management strategies to control and eliminate PRRSv from infected pig populations. Specifically, immunological interventions were evaluated to: a) determine the time it takes to reach PRRSv-stability on infected breeding herds undergoing PRRSv control and elimination program b) assess the impact of immunological interventions on production parameters, c) identify risk factors associated with successful PRRSv elimination protocols and, b) reduce PRRSv shedding from infected growing pig populations.

A better understanding of effectiveness of immunological interventions will allow producers and veterinarians to make informed decisions to control and eliminate PRRSv from infected pig herds, systems and ultimately from geographical regions.

Chapter I. Literature review

Etiology

Porcine reproductive and respiratory syndrome (PRRS) was first denominated "mystery swine disease", "porcine epidemic abortion and respiratory syndrome", "blue-ear pig disease" and "swine infertility and respiratory syndrome (SIRS)". The causative agent is PRRS virus (PRRSv). PRRSv is also considered a major component of the porcine respiratory disease complex (PRDC) and, more recently, it has been associated with epidemics in China called "high fever pig disease" (Zhou and Yang, 2010).

PRRSv is a 50-65 nm of diameter single-stranded positive-sense enveloped RNA virus belonging to the Nidovirales order, Arteriviridae family and Arterivirus genus, which also includes lactate-dehydrogenase elevating virus (LDHv), equine arteritis virus (EAv) and simian hemorrhagic fever virus (SHFv) (Benfield et al., 1992; Cavanagh, 1997; Meulenberg et al., 1993). PRRSv has a ~15Kb genome consisting of at least 7 open reading frames (ORFs), 1 (1a and 1b), 2 (2a and 2b), 3, 4, 5, 6 and 7. ORFs 1a and 1b comprise 75% of the viral genome and code for proteins associated with viral replication (Meulenberg et al., 1993; Snijder and Meulenberg, 1998). ORFs 2-7 code for viral structural proteins named GP2, GP3, GP4, GP5, M and N proteins respectively. Recently ORF 5 was sub-divided into 5a and 5b regions, coding for GP5 and ORF5a proteins (Johnson et al., 2011). Proteins N, M and GP5 are considered to be the main structural proteins and are required for viral particle formation and infectivity, while GP2, GP3 and GP4 are considered minor structural proteins and are required for viral infectivity (Wissink et al., 2005).

There are two main PRRSv genotypes; the European (also known as “Lelystad” or type 1 PRRSv) (Wensvoort et al., 1991) and the North American genotype (VR-2332 or type 2 PRRSv) (Benfield et al., 1992; Collins et al., 1992). These two main genotypes differ genetically by approximately 40% (Meulenberg et al., 1993). Within each genotype there is nucleotide divergence of up to 30% (Murtaugh et al., 2010; Nelsen et al., 1999; Shi et al., 2010a). Koch’s postulates have been fulfilled with both genotypes (Collins et al., 1992; Terpstra et al., 1991; Wensvoort et al., 1991).

Pathogenesis and clinical manifestation

The economic impact of PRRSv is significant and in the U.S alone it was estimated at $74.16/litter in affected sow farms and $6.01 and $7.67 per pig in the nursery and grower-finisher phases respectively (Neumann et al., 2005). A recent study estimated that PRRS costs to the US swine industry over 1 billion dollars annually (NPB, 2011).

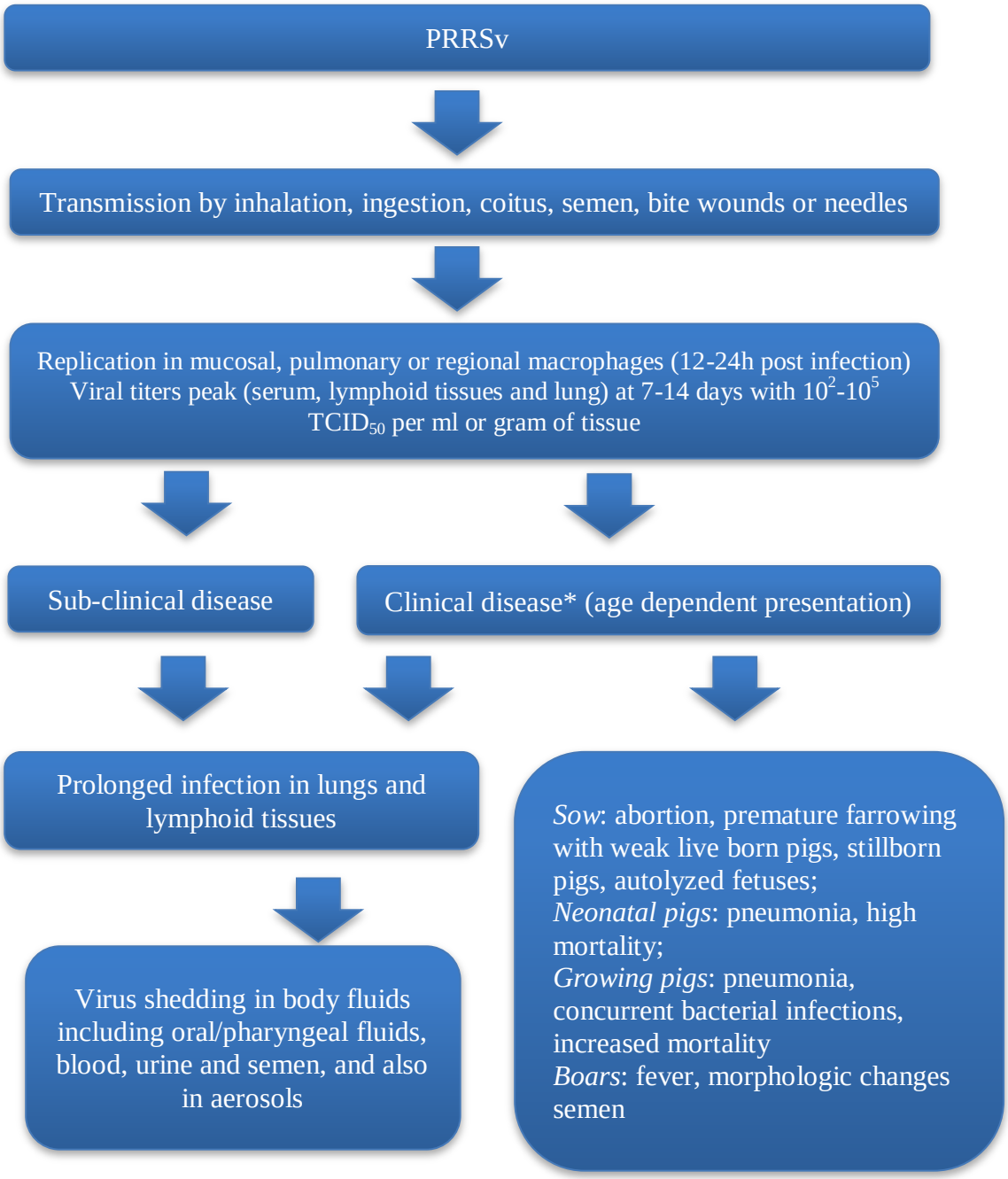
PRRSv produces a viremic infection which lasts approximately 3-5 weeks (Batista et al., 2004; Xiao et al., 2004) and a non-viremic prolonged infection of lungs and lymphoid tissues that can last up to 5 months post infection (Allende et al., 2000; Wills et al., 1997c). One study reported PRRSv RNA detection by RT-PCR in serum and tonsil samples at 225 and 251 days post experimental infection, respectively (Wills et al., 2003).

When PRRSv infects a susceptible host it primarily targets monocyte/macrophage lineage cells for replication. The PRRSv pathogenesis is summarized in figure 1.

Clinical manifestation of PRRS varies depending on herd immunity levels, pig susceptibility and PRRSv strain. PRRSv can range from unapparent infection with minimal impact in productivity to severe reproductive and/or respiratory disease (Zimmerman et al., 1997). The duration of herd epidemics is expected to be longer in infected herds with partial immunity as compared to naïve herds (Pulliam et al., 2007). As pigs grow, they become less susceptible to developing characteristic clinical signs of PRRSv infection (Klinge et al., 2009; Thanawongnuwech et al., 1998; van der Linden et al., 2003; Zimmerman et al., 1997). However, all pigs develop viremia and antibody response. Common PRRS clinical signs in growing pigs include lethargy, anorexia, hyperemia (39-41ºC), rough hair coats and variable worsening of growing performance creating uneven groups (Keffaber, 1989; Loula, 1991). Affected growing pigs become more susceptible to secondary infections including *Salmonella choleraesuis* and *Mycoplasma hyopneumoniae* in pre-weaned and weaned pigs (Christianson et al., 1992; Keffaber, 1989; Loula, 1991). Because of the loss of appetite and fever, sows have increased risk to develop agalactia, which leads to decreased colostrum and milk intake by piglets which become more prone to secondary diseases (Christianson et al., 1992; Thacker, 2003).

PRRSv strain variability plays an important role in the manifestation of PRRS clinical signs. Susceptible pigs infected with “highly virulent” PRRSv strains, as compared to low virulent strains, develop higher viral load in lymphoid tissues (Halbur et al., 1996) and viremia of higher duration and magnitude (Grebennikova et al., 2004). There are reports of PRRSv “high virulent” strains producing severe clinical signs in

piglets and sows (Hurd et al., 2001; Mengeling et al., 1998; Rossow et al., 1999; Thacker, 2003).



* Apoptosis and death of infected and proximate non-infected cells, induction of inflammatory cytokines, induction of polyclonal B cell activation and reduction in bacterial phagocytosis and killing by macrophages.

Figure 1 - Pathogenesis of PRRSv infection. Adapted from Rossow et al (Rossow, 1998) and Zimmerman et al (Zimmerman et al., 2006).

Diagnosis

Clinical signs and serology

Analysis of production parameters and clinical signs can lead to suspicion of PRRS. However, laboratorial diagnosis is required for confirmation of the involvement of PRRSv. Presence of anti-PRRSv antibodies in serum or plasma samples detected by enzyme-linked immunosorbent assay (ELISA) (i.e. IDEXX PRRS 2XR Ab test, IDEXX Laboratories, Inc., Westbrook, Maine) confirms PRRSv infection from 1-3 weeks up to 12-24 months post infection. It has been reported that IDEXX ELISA test specificity is not 100% (Torremorell et al., 2002) and therefore positive results need to be confirmed by indirect fluorescent antibody (IFA) or immunoperoxidase monolayer assay (IPMA) test in PRRSv-negative populations.

Molecular diagnosis

Reverse transcriptase polymerase chain reaction (RT-PCR)-based assays are used to detect PRRSv RNA in samples (Christopher-Hennings et al., 1995). The sample of choice depends on stage of infection. Serum samples are the preferred sample for diagnosing recently infected pigs, and lung and lymphoid tissue samples (tonsils and systemic lymph nodes) are preferred for pigs in the prolonged infection stage. PCR-based assays can also be used for estimating shedding status (semen, oral fluids and aerosol samples) of pigs. Furthermore, differentiation between PRRSv isolates is

performed by phylogenetic comparison of PRRSv open reading frame 5 (ORF 5) nucleotide sequences.

Determination of PRRSv infectiveness

To assess whether PRRSv present in samples is infectious, the virus needs to be isolated in cell cultures (such as porcine alveolar macrophage or MARC-145 cell lines) followed by either fluorescent antibody or immunohistochemistry staining of cells to detect viral antigens (Benfield et al., 1992). However, for maximal sensitivity the swine bioassay method is recommended (Horter et al., 2002). It consists of injecting sample homogenate into a naïve sentinel pig and observing the recipient pig to seroconvert and develop PRRSv viremia. A 81% agreement between positive PCR results and detection by swine bioassay of infectious PRRSv in oropharyngeal scraping samples has been reported (Horter et al., 2002).

The differential diagnoses for PRRS clinical signs include infectious agents such as parvovirus, pseudorabies virus, hemagglutinatting encephalomyelitis virus, porcine circovirus type 2, enterovirus, influenza virus, classical swine fever virus, cytomegalovirus and leptospirosis (Zimmerman et al., 2006).

Epidemiology

Transmission

PRRSv is a highly infectious (Yoon et al., 1999) but not necessarily a highly contagious (Potter, 1994; Zimmerman, 2003) pathogenic agent. PRRSv can be found

in lungs and lymphoid tissues as well as in different body fluids, secretions and excretions including urine, blood, tracheal rinse, oral fluids (Wills et al., 1997a), nasal secretions (Rossow et al., 1994), semen (Swenson et al., 1994), feces (Christianson et al., 1993), mammary secretions (Wagstrom et al., 2001) and aerosols (Cho et al., 2006b). Therefore, direct contact with PRRSv-infected pigs is a major route of infection. It has been estimated that PRRSv at concentrations of 10^1 $TCID_{50}$ or higher is sufficient to cause infection in susceptible pigs via the intranasal or intramuscular route (Yoon et al., 1999). Indirect contact is also an important route of infection and includes PRRSv-contaminated boots, coveralls and needles (Otake et al., 2002). PRRSv can also be transmitted from herds via insects that act as mechanical vectors (Otake et al., 2004). Contaminated transport trailers represent another important route of area dissemination (Dee et al., 2004c). There are validated methods for trailer disinfection (Dee et al., 2004a, 2005a; Dee et al., 2005b) although those demand considerable fixed and variable costs and depend on proper compliance of standard protocols. Finally, considering that the introduction of PRRSv-carrying aerosols in pig farms cannot be prevented unless the farm is air-filtered, aerosols constitute one of the most important routes of viral regional dissemination in high dense areas (Desrosiers, 2011; Mortensen et al., 2002). Infectious PRRSv has been detected in air samples collected to up to 9.1 km from an endemically infected growing-pig population (Otake et al., 2010). Aerosols of a PRRSv strain MN-184 at dose of $10^{0.26}$ $TCID_{50}$ were sufficient to cause airborne infection in susceptible piglets (Cutler et al., 2011).

Risk factors associated with PRRSv transmission and clinical signs

Mortensen and colleagues studied the risk factors associated with infection with vaccine associated-PRRSv strains. The significant risk factors were proximity to PRRSv-infected herds, purchase of pigs or semen from PRRSv-infected sources and increasing herd size (Mortensen et al., 2002). Others also reported that herds neighbored by PRRSv-positive sites had higher risk of infection (Le Potier et al., 1997; Velasova et al., 2012).

A higher "external score" from version 1 of the Production Animal Disease Risk Assessment Program (PADRAP, www.padrap.org) as well as winter months were associated with infection of breeding herds (Holtkamp et al., 2010). The second version of the PADRAP risk assessment for breeding herds was published identifying 89 risk factors associated with higher risk of PRRSv infection, confirming the findings of the first version and including other variables such as type of breeding herd (commercial herds had a higher risk compared to multipliers), entry of replacement animals into the breeding herd and breeding female per on-site employee (Holtkamp et al., 2012). That same study reported that breeding herds had higher risk of PRRSv infection when the semen source site (boar stud) had recent history of PRRSv infection or was located in a high pig dense region.

Dee and colleagues reported that the meteorological risk factors associated with the presence of PRRSv in air are cool temperatures, wind direction, low sunlight levels, winds of low velocity in conjunction with gusts and rising humidity and atmospheric pressure (Dee et al., 2010a). Neumann et al reported a model describing the risk of

PRRSv introduction in PRRSv-free swine industries by importation of virus-contaminated pork (Neumann et al., 2007).

Goldberg and colleagues reported that severity of PRRS clinical manifestation is associated with increasing herd size, PRRSv strain and management practices including all in-all out (AI-AO) practices (Goldberg et al., 2000). Similarly, Young et al reported that severity of PRRS clinical signs are associated with increasing herd size (Young et al., 2010). Moreover, it has been reported that farms were more likely to have active PRRSv infection if they had dead pigs collected by an external rendering provider (compared to sites that composted carcasses) and less likely to have PRRSv infection when the average age of weaned pigs was superior to 27 days of age or when an attenuated vaccine to control PRRSv infection was used (Velasova et al., 2012).

PRRSv regional dissemination

As reviewed elsewhere PRRSv can be transmitted between herds directly via transport of infected pigs or semen, as well as via indirect routes including contaminated vehicles, fomites and insects as mechanical vectors (Cho and Dee, 2006). Moreover, airborne transmission is also a major route of PRRSv regional spread (Desrosiers, 2011).

Control and elimination

There is no drug to stop PRRSv replication in the host. Therefore, pig herds need to adopt strict biosecurity programs (Julio Pinto and Santiago Urcelay, 2003; Pitkin et

al., 2009c; Spronk et al., 2010) to prevent PRRSv infection. Once infected, the options are to: a) keep the farm PRRSv positive and control manifestation of clinical disease or b) eliminate PRRSv from the farm. To eliminate PRRSv from a pig farm, there are different strategies including: a) complete herd depopulation and repopulation with PRRSv-negative pigs (Dee, 2003), b) test and removal (Dee and Molitor, 1998), c) herd closure (Torremorell et al., 2003) and d) partial depopulation (Dee et al., 1997a; Dee et al., 1997b).

Control strategies

When PRRSv elimination is not possible, farms adopt strategies to control viral shedding and viral spread. The goal of control strategies for PRRSv in breeding herds is to produce PRRSv-negative piglets at weaning (Corzo et al., 2010), and minimize PRRSv shedding and consequent horizontal transmission in growing pigs. The documented strategies to control PRRSv are:

1 - Proper gilt acclimatization (Dee, 2003; Dee et al., 1995) - which entails that gilts must have already developed protective immunity to PRRSv prior to their introduction as adults in the breeding herd (Torremorell et al., 2002). To accomplish this, gilts must be exposed to PRRSv at an early age around 12 weeks of age, or at least 2-3 months before introduction into the breeding herd (FitzSimmons and Daniels, 2003). This method can also be used as part of programs to eliminate PRRSv from breeding herds (Fano et al., 2005).

2. - Partial depopulation methods – consist of depopulating the nursery (weaning to an off-site location) in order to break the cycle of PRRSv transmission in production systems producing PRRSv negative offspring (Dee and Joo, 1994; Dee et al., 1997b). The acknowledged limitations of this method include the logistics to wean piglets to an offsite location and the necessity to repeat the method every 1-2 years.

3. - Management practices to minimize PRRSv horizontal transmission – includes (a) all in-all out (AI-AO) pig flow, consisting of preventing infection of younger pigs by older ones (Dee, 2003); (b) application of McREBEL, which includes management changes to reduce exposure to bacteria to eliminate losses (McCaw, 2000); and (c) adoption of biosecurity protocols (Pitkin et al., 2009c).

4. - Immunization of pigs – consists of inoculating PRRSv in pigs with an attempt to build herd immunity against the virus (Dee and Phillips, 1999; Lowe et al., 2006). The resident live virus (aka serum inoculation), attenuated live virus vaccines and/or killed virus vaccines can be used as immunogens (Fano et al., 2005; Lowe et al., 2006). As opposed to the former two methods, it appears that application of killed PRRSv vaccines does not elicit protective immunity in piglets when used as sole immunization method (Zuckermann et al., 2007). A recent study to evaluate the effect of killed autogenous vaccines on PRRSv infection dynamics concluded that “the experimental heterologous inactivated vaccines and the commercial inactivated vaccine had no or only a limited influence on viremia” (Geldhof et al., 2012). Moreover, it has been hypothesized that a live PRRSv replication is required to provide immunological protection against PRRSV infection (Murtaugh and Genzow, 2011). Thus, under field conditions in the US swine industry veterinarians have

reported use of a live PRRSv (resident live virus or attenuated vaccine) for immunological treatment of infected herds (Desrosiers and Boutin, 2002; Heller et al., 2011; Lowe et al., 2006; Voglmayr et al., 2006). However to our knowledge there is no report on the comparative effectiveness of different types of live PRRSv (resident virus vs MLV vaccine) on the ability to control and/or eliminate PRRSv from infected pig populations.

5. - Temporal herd closure – discontinuation of the introduction of susceptible pigs for 4-8 months with an attempt to break the PRRSv transmission cycle within the herd (Dee et al., 1995; Torremorell et al., 2003).

6. - Vaccination and unidirectional flow – PRRSv live attenuated vaccine associated with unidirectional flow is used in growing pig populations to control and potentially eliminate PRRSv from segregated finishing populations (Dee and Phillips, 1998).

Elimination strategies

Unless eliminated, PRRSv infection can last forever in a herd due to the constant introduction of new groups of pigs and presence of pigs at various stages of PRRSv-infection and immunity (Dee et al., 1996; Dee et al., 1997a). Therefore, several strategies to eliminate PRRSv from production systems have been documented. A PRRSv-negative site is defined as a herd with pigs consistently testing negative for ELISA and PCR-based tests using 30 samples/month over a 90 day period and absence of PRRS clinical signs (Holtkamp et al., 2011).

Different methods have been reported to eliminate PRRSv from pig production systems, including:

1. - Whole herd depopulation-repopulation – as the name suggests, it consists of depopulating the whole breeding herd and repopulating it with PRRSv negative pigs after the farm has been cleaned and disinfected (Dee, 2003).

2.- Test-and-removal – After sampling the whole breeding herd (breeding herds or boar studs) and testing using PRRSv diagnostics (serology and molecular diagnostics), PRRSv-positive pigs are removed from the herd (Dee and Molitor, 1998).

3. - Partial depopulation – as described for the PRRSv control section.

4. - Segregated early weaning – piglets are weaned at 8-14 days of age to an off-site location. Upon confirmation of pigs being anti-PRRSv antibody free (ELISA tests) and after the decrease of maternal antibodies (around 10 weeks of age), pigs are considered free of PRRSv (Rajic et al., 2001). This method requires that the sows of the source farm are not transmitting PRRSv vertically to the fetuses and not shedding virus to the newborns.

5 - Herd closure – Consists of stopping introduction of replacement pigs until herd immunity is established and negative offspring are consistently produced (Torremorell et al., 2003). Herd closure is financially advantageous over complete herd repopulation (Holtkamp et al., 2012; Zimmerman et al., 2006) and it is commonly associated with deliberate exposure of all animals of the breeding herd to a live virus (Desrosiers and Boutin, 2002; Voglmayr et al., 2006). Thus, the entire pig

population is infected and eventually develops a protective immune response to PRRSv, which results in herd immunity and elimination of PRRSv from the breeding herd.

Concept of regional elimination of swine pathogens

As previously discussed, PRRSv-free farms can become infected via various routes and they represent a risk to other farms in the region. For the purpose of this thesis, a region is considered a "pre-defined geographic area with clearly defined limits"[1]. The word dissemination comes from the latin *disseminatus* (dis – "in all directions" + seminare – "propagate"). The definition, example of usage and synonyms below were extracted from the Merriam-Webster's third new international dictionary (2002):

> Dissemination - To make widespread; to disperse throughout small particles; to distribute in every part; to extend widely; to scatter over a large area or into many places.
>
> "Distrusting the great city twenty miles away that disseminated its virus through the outlying villages and farms".
>
> Synonyms: spread, diffuse, permeate.

Eliminating a pathogen from a region obviously decreases dramatically the risk of pathogen introduction in free herds of that specific region. The concept of regional elimination of swine pathogens is not new. In fact, it has been well documented. For instance, swine vesicular disease virus has been eradicated from Italy (Bellini et al.,

[1] Merriam-Webster's third new international dictionary. 2002.

2007), foot and mouth disease virus has been eradicated from many countries including Colombia (Gallego et al., 2007), Philippines (Randolph et al., 2002) and Japan (Sugiura et al., 2001). Similarly, Aujeszky's disease virus has been eradicated from many regions including Spain (Allepuz et al., 2008), New Zealand (MacDiarmid, 2000) and Netherlands (Elbers et al., 2000). Also, African swine fever virus has been eradicated from Brazil (Moura et al., 2010).

The knowledge on how to control and eliminate PRRSv from individual pig herds as well as the basic understanding of PRRSv regional dissemination routes allowed the viability of regional PRRSv elimination projects. Chile was the first country that successfully eliminated PRRSv from its entirety (Torremorell et al., 2008). PRRSv has also been eliminated from Sweden (Carlsson et al., 2009). There is also report of efforts to eliminate PRRSv from the "Pays de Loire" region in France (Le Potier et al., 1997). In the United States, regional PRRSv elimination has been reported for the Stevens county, Minnesota, which consists of 1,490 Km2 with 87 pig sites (Corzo et al., 2010).

Regional pathogen elimination projects can happen as voluntary or mandatory programs. In some instances as it was the case of Aujeszky's disease virus elimination from the US swine industry started as a volunteer-based program and ended up as a mandatory program. Coordinating a volunteer-based project,(Corzo et al., 2010; Rowland and Morrison, 2012) requires (a) leadership to unite as many producers and veterinarians in the ultimate project goal; (b) constant communication and meetings to show project progress and recruit new participants; (c) understanding of the region's pig flow; (d) record premises characteristics, herd density and PRRSv

status of each site in the region; (e) have the region's boundaries well defined; and (f) continuous surveillance to ensure that farms that adopted virus elimination strategies remain PRRSv negative. All of this requires (d) funding source(s). There are more than 20 regional elimination projects ongoing in the US swine industry at this time and those are financially supported by USDA-funded projects and private industry (Menard, 2012).

Once PRRSv has been eliminated from specific farms, those face the challenge to remain negative despite the risk of re-infection. To remain PRRSv-free, farms adopt strict biosecurity measures, which include use of new technologies such as air filtration to prevent airborne infection (Dee et al., 2010b). Ideally, farms that successfully eliminated the virus would continue monitoring the herd periodically and report to the regional elimination project leader when re-infection occurred. However, in some instances this active surveillance approach is not feasible. Therefore, a risk-based surveillance approach is an option where the project leader facilitates testing of pig herds at relatively highest risk of infection (Rowland and Morrison, 2012).

It has been documented that one of the main challenges of regional PRRSv elimination projects in the U.S. swine industry is the re-infection of farms that attempted virus elimination, specially those located in high pig dense regions due to regional spread of the virus via many different routes (Corzo et al., 2010; Holtkamp et al., 2010). Therefore, there is an urgent need to validate cost-effective field strategies to minimize PRRSv regional dissemination at the different stages of pig production.

Summary

Much has been learned about PRRS etiology and how to diagnose, control and eliminate PRRSv from infected herds. Methods to prevent new infections through biosecurity improvements have also been learned.

Herd closure is a widely practiced method in the US swine industry to control and eliminate PRRSv from breeding herds. However, failure to eliminate virus using this method has been reported (Torremorell et al., 2003). There is a need for studies to better understand the science behind herd closure, and what are the herd closure-associated practices correlated with success. Some of the unanswered questions are: which method of whole herd exposure is more efficient? What role does number of whole-herd exposures play on effectiveness of herd closure programs? Does prior immunity to PRRSv interfere with herd closure success to eliminate the virus?

Furthermore, it has been reported that PRRSv-negative growing pig populations frequently get infected with PRRSv especially in high pig dense regions (Loula, 2011). One could hypothesize that those infections happen due to the high PRRSv pressure of infection. In other words, there is a considerable amount of PRRSv being disseminated from infected pig sites, which expose surrounding PRRSv-negative sites. Therefore, there is a need for studies to evaluate strategies to minimize risk of PRRSv dissemination from infected pig populations.

Chapter II - Comparison of time to PRRSv-stability from infected breeding herds between load-close-expose programs

Linhares DCL, Cano JP, Torremorell M, Morrison RB. Comparison of time to produce PRRSv-stability from infected breeding herd between load-close-expose programs. [in prep, to be submitted to Prev Vet Med].

Abstract

Since its first report in the swine industry, porcine reproductive and respiratory syndrome virus (PRRSv) continues to cause significant pig production losses around the world. Herd closure combined with whole herd exposure to a live PRRSv is a method commonly used to control and eliminate PRRSv from breeding herds. This strategy consists of interrupting introduction of replacement pigs in the herd and immunizing the breeding herd by inoculation of a replicating PRRSv. This was a prospective study that followed 61 breeding herds acutely infected with PRRSv that adopted herd closure and whole herd exposure with either a modified live virus (MLV) vaccine or live-resident virus inoculation (LVI) aiming to produce PRRSv-negative pigs at weaning. Time to PRRSv stability (TTS) was compared between MLV and LVI herds using Kaplan-Meier survival curves. Multivariate analysis using proportional hazards regression was performed adjusting the effect of treatment on TTS by selected herd characteristics. Day 1 of the program was considered to be the day that treatment (LVI or MLV) was administered. Sampling at the herds started at 12 weeks after day 1 and consisted of bleeding 30 due-to-wean piglets on a monthly basis. Serum was tested for PRRSv RNA by RT-PCR. Herds in which PRRSv was not detected in sampled pigs over a 90-day period were classified as reaching TTS. The median TTS among participating herds was 26.6 weeks (25th to 75th percentile, 21.6 to 33.0 weeks). Twenty-four herds had at least 1 month of PRRSv PCR-negative results followed by positive PCR results indicating that PRRSv monitoring in herds undergoing virus elimination must be done repeatedly over time. The frequency of having at least one PCR negative result followed by a positive one tended to be higher in the MLV group. Farms reached TTS significantly sooner when they a) were

treated with LVI, b) had history of PRRSv infection in the previous 3 years of the study and c) were assisted by a specific veterinary clinic. This study provided important information to assist veterinarians decide between methods of whole-herd exposure to produce PRRSv-negative pigs from infected breeding herds.

Keywords: PRRSv, control, elimination, load-close-expose, whole herd exposure, live virus inoculation, serum inoculation, modified-live virus vaccine.

Introduction

Since its first report in the US swine industry 23 years ago (Keffaber, 1989), porcine reproductive and respiratory syndrome virus (PRRSv) continues to cause significant pig production losses in North America and around the world (Lunney et al., 2010; Zimmerman et al., 2012). There are different methods to control and eliminate PRRSv from swine breeding herds, including whole herd repopulation, test and removal, and herd closure (Corzo et al., 2010).

A standardized terminology to communicate PRRSv-status for breeding herds has been proposed. Briefly, PRRSv-infected herds can be classified as "positive unstable". Herds undergoing PRRSv elimination intend to move to "positive stable" status, and then to "provisional negative" and ultimately to "negative" status. (Holtkamp et al., 2011). Shedding of PRRSv can be expected from stable herds, whereas provisional negative and negative herds should have pigs testing negative on PCR-based testing for PRRSv RNA. Provisional negative herds have pigs that potentially have anti-PRRSv antibodies (i.e. "ELISA-positive"), whereas all pigs from "negative" herds are free from both PRRSv and anti-PRRSv antibodies (Holtkamp et al., 2012).

Herd closure is a strategy commonly used with the goal to eliminate PRRSv from infected herds and is financially advantageous over whole herd depopulation because there is no required down-time, sows are not slaughtered, and there is no clean-up cost (Torremorell et al., 2003). Herd closure consists of interrupting the introduction of replacement pigs in the sow herd for several months, while remaining females are continuously bred and thus production is not ceased. Herd closure success rate was

estimated at above 85% for farms with segregated production (Torremorell et al., 2003).

Torremorell and colleagues first described herd closure programs as a method to control and/or eliminate PRRSv from herds endemically infected. Those breeding herds had limited clinical signs and low preweaning piglet mortality rate (Torremorell et al., 2003). In recent years, veterinarians have extended the use of herd closure protocols to acutely infected herds (Desrosiers and Boutin, 2002). Under these conditions, there is less herd immunity in the population.

As part of the herd closure program veterinarians may advocate deliberate exposure with PRRSv of all breeding pigs in the herd in an effort to increase herd immunity (Desrosiers and Boutin, 2002; Voglmayr et al., 2006). In conjunction with deliberate exposure, the program includes filling the facility with extra replacement pigs and then temporarily closing the breeding herd to any introduction of replacement pigs. Hence, the herd closure program is commonly thought of as having “load, close and expose” components.

There are different immunogens that can be used for whole-herd exposure including resident (virulent) live virus and modified-live virus (MLV) vaccines. The term “resident virus”, also referred as “wild type” or “field virus” has been used to describe the virus detected at a specific pig farm. Attenuated vaccines, in turn, are commercially available, USDA licensed products commercially available. To our knowledge, there is no scientific data comparing the effectiveness of different immunization protocols to reach PRRSv-stable status 66from PRRSv-positive sources under field conditions.

It has been estimated a PRRSv-positive growing pig has an additional production cost of $13.52 (Neumann et al., 2005). Therefore, one way to compare PRRSv elimination programs in breeding herds is to measure the time it takes to produce PRRSv-negative offspring from infected breeding animals. The purpose of this study was to compare the time required to PRRSv-stability at weaning between breeding herds that used MLV or live-virus inoculation (LVI) as immunogens as part of the load-close-expose programs.

Design and methods

Study design

We conducted a prospective quasi-experimental study to compare time-to-PRRSv-stability (TTS) from PRRSv-positive breeding herds using either MLV vaccine or LVI for whole-herd exposure as part of a load-close-expose program. In the MLV group, herds were exposed to either Ingelvac PRRS® ATP (Boehringer Ingelheim Vetmedica Inc, St. Joseph, MO, USA) vaccine or Ingelvac PRRS® MLV (Boehringer Ingelheim Vetmedica Inc., St. Joseph, MO, USA) vaccine. LVI protocols were based on inoculating diluted PRRSv PCR-positive serum collected from infected pigs from a specific herd (Batista et al., 2002).

Herds were assumed to achieve “TTS status” when there was a failure to detect PRRSv RNA in serum of pre-weaning pigs by RT-PCR over a 90-day period. Day 1 of the program was the day that the herd was exposed to either MLV or LVI.

Herds were dropped from the study if (a) the herd veterinarian indicated that there was a new unrelated PRRSv introduction, (b) the participating herd had introduced replacement animals before TTS was reached or (c) the owner opted to withdraw.

The study was approved by the University of Minnesota's IACUC (protocol number 1004A80315) and all the farm owners or their representatives provided written, informed consent agreeing to participate.

Source population

Veterinarians listed on the American Association of Swine Veterinarians (AASV) directory were invited via email to enroll herds to which they provided technical services. Eligible herds were those acutely infected with PRRSv from November 2009 to March 2012 and that adopted load-close-expose program with intent to eliminate PRRSv from the herd. Enrolled herds fulfilled the following inclusion criteria: a) farrow to wean breeding herd; b) diagnostic evidence that PRRSv was present; c) no diagnostic evidence that the herd had been infected during the closure period with a previously undetected PRRSv isolate; d) intent of herd owner to eliminate PRRSv from the herd; e) no replacement pigs introduced during the closure period; f) owners willing to cover costs of collecting samples and diagnostic testing as defined by the project and g) enrollment in the study before the PRRSv monitoring phase started. Herds that did not fulfill these criteria were not considered in the analysis for this study.

A summary of the baseline demographic characteristics of the enrolled herds can be seen in Table 1. A total of 61 herds were enrolled and information on herd demographics, type of PRRSv strain and PRRSv infection history of the herd were recorded.

Monitoring

PRRSv-monitoring started 12 weeks after day 1 and consisted of bleeding 30 pre-weaned piglets (determined by target prevalence to detect at least one positive assuming a 10% prevalence at a 95% confidence level for any population size) and testing serum samples for PRRSv-RNA by RT-PCR. Blood serum samples were submitted to a reference veterinary diagnostic laboratory for testing in pools of 5 (Rovira et al., 2007) following the standardized procedures for each laboratory.

Farms that obtained intermittent PCR-positive results (e.g. two months of negative results followed by a positive result) submitted the serum samples for PRRSv RNA open reading frame 5 (ORF 5) molecular sequencing. The sequences were compared to the resident virus (phylogeny comparison of the PRRSv-ORF5 previously detected in the herd). For the purpose of this study, unexpected PCR-results were considered to be the same resident virus (no new PRRSv introduction) when there was (a) no apparent increase in clinical signs, and (b) the sequenced ORF 5 was at least 97% similar to the resident virus present at the beginning of the load-close-expose program (Alonso, 2012; Murtaugh, 2012).

Outcomes

Effectiveness of load-close-expose programs was compared between treatment groups using TTS as the outcome parameter.

The effect of multiple covariates on TTS and their association with the main effect (MLV or LVI treatment) was assessed. The following covariates were considered in the analysis: herd size, previous PRRSv infection, PRRSv restriction fragment length polymorphism (RFLP) pattern 1-4-4, period of time between PRRSv-detection and whole-herd exposure and veterinary clinic that provided technical services.

Herds that were PRRSv-free (AASV category IV) (Holtkamp et al., 2011) for the previous 3 years before PRRSv infection were considered to have “no prior PRRSv infection”. Herds with at least one PRRSv introduction in the past 3 years before the current infection were considered to have “PRRSv prior infection”. A dichotomous variable “RFLP pattern 1-4-4” was created to adjust analysis for classifying PRRSv belonging to a similar virus genotyping group (Shi et al., 2010b).

Statistical analysis

Power calculation indicated that a minimum of 31 and 15 herds for the LVI and MLV groups respectively were needed to have at least 80% power to detect differences between treatments, assuming a hazard ratio of 2.50, with the use of a log-rank test at a two-sided significance level of 0.05. Overall, a higher number of herds in the LVI group were enrolled because the number of eligible herds using MLV vaccine in load-close-expose programs was proportionally fewer than that of LVI-herds.

Treatment groups (MLV or LVI) were compared with the use of Kaplan-Meier survival estimates for TTS, with censoring data for herds that dropped from the study after PRRSv monitoring started (12 weeks) and before reaching TTS. Herds that dropped from the study before starting PRRSv monitoring were excluded from the analysis. Between group differences in survival curves were assessed with the use of log-rank test.

Cox proportional-hazards regression was used to calculate the magnitude of the treatment effect on TTS and to adjust the analysis for the aforementioned covariates. Factors associated with TTS with a p-value less than 0.20 in the univariate analysis were entered in the multivariate model and non-significant factors were removed by

means of a backward-selection procedure. Two-sided hypotheses and tests were used for all statistical inferences. A p-value of less than 0.05 was considered to indicate statistical significance.

Statistical analyses were performed with the use of SAS 9.2 software (SAS Institute Inc., Cary, North Carolina).

Results

Descriptive analyses

Eighty-five percent of herds in the LVI program and 60% of herds in the MLV program reached TTS by the end of the observational period (table 2).

TTS ranged from 12 to 42 weeks among 61 participating herds with a median of 26.6 weeks (25th to 75th percentile, 21.6 to 33.0 weeks). From 61 farms, 24 had at least 1 month of PRRSv PCR-negative results followed by positive PCR results (Table 3). Overall, frequency of having at least one PCR negative result followed by a positive one tended to be higher in the MLV group (Chi-square p-value 0.0804).

In contrast to MLV group, the LVI group had no event of 2 or 3 negative tests followed by a positive result (Table 4). Farms that had at least two consecutive PCR-negative followed by PCR-positive results obtained sequences at least 97% similar to the original resident virus, suggesting that those unexpected PCR-positive were not explained by lateral PRRSv infection (Christopher-Hennings et al., 2002; Murtaugh, 2012).

Univariate analyses

Herds that were exposed with LVI had significantly shorter TTS than herds treated with MLV (Log rank p-value 0.0137) (Figure 1A). Similarly, herds with prior PRRSv infection reached TTS significantly sooner than herds without PRRSv prior-infection (Log rank p-value 0.0275) (Figure 1B).

RFLP pattern 1-4-4, herd size and time from PRRSv-detection to whole-herd exposure were not statistically associated with TTS (Table 4).

Multivariate analysis

The variables that remained in the final multivariate model were treatment (LVI or MLV) and prior PRRSv-infection (yes or no). The following variables were not associated with TTS: herd size, RFLP pattern and days from PRRSv-detection to whole-herd exposure.

After adjusting for treatment and prior PRRSv-infection status, herds assisted by a specific veterinary clinic achieved TTS significantly sooner than all other herds (p-value 0.0429).

The final Cox regression model hazard ratio for TTS was 2.9 for LVI herds compared to MLV herds (p-value 0.0034) and 2.5 for herds with prior PRRSv infection compared to herds without prior-PRRSv infection (p-value 0.0036).

Discussion

The purpose of the present study was to investigate whether type of immunogen (LVI or MLV) used in load-close-expose programs to eliminate PRRSv from breeding herds influenced the time required to PRRSv-stability. TTS can be defined as a proxy

of time to produce PRRSv-negative pigs at weaning. However, as the sample size used to monitor PRRSv status of herds was 30 pigs, there was no statistical power to detect PRRSv-positive PCRs when prevalence at piglet population was below 10%.

Overall our results indicated that after an acute outbreak the time it took to reach PRRSv stability following adoption of load-close-expose was variable when compared among herds of similar demographics. Further studies are needed to better understand risk factors associated with herds that achieved shorter TTS.

The median TTS of 26.6 weeks was longer than we anticipated given previous reports in the literature (Rowland and Morrison, 2012; Torremorell, 2011; Yeske, 2009; Zimmerman et al., 2012). Perhaps the previous reports were based on a few "success" cases and did not fully represent the range of herds represented in this study. To our knowledge, this is the first study to prospectively follow a relatively high number of PRRSv infected herds attempting to eliminate the virus and thus we speculate that the median TTS reported in this study could be extrapolated to other herds that have characteristics similar to the herds enrolled in this study (described in the eligibility criteria list in materials and methods).

Follow-up research is needed to identify herd-level risk factors that may explain the wide variability in TTS and the frequency of intermittent detection of PRRSv-RNA by PCR. Enforcement of management practices to reduce within-herd transmission of pathogens (McCaw, 2000) could be associated with shorter PRRSv-TTS. The fact that herds assisted by a specific veterinary clinic achieved TTS sooner might be explained by differences in standard protocols adopted by such herds during the herd closure period. In addition, data from this study proves that during PRRSv

elimination programs PRRSv monitoring must be done repeatedly over time (Table 4) as previously suggested (Holtkamp et al., 2011).

The adjusted comparisons showed that after adjusting for prior PRRSv infection status, herds of the LVI group had a higher TTS hazard ratio than those of MLV group. Given that a particular herd did not reach TTS yet, it had a 2.9 relative risk (hazard ratio) of reaching TTS in the following week.

There are two fundamental differences between exposure programs using LVI and MLV that could explain why LVI herds achieved TTS faster than MLV herds. First, immune response to virulent PRRSv is faster and stronger than that to attenuated virus, as measured by duration and magnitude of viremia and antibody titers (Klinge et al., 2009). Secondly, even though the degree of RNA nucleotide similarities between viruses does not predict degree of immunological cross-protection (Murtaugh and Genzow, 2011; Prieto et al., 2008), exposing the breeding herd to resident virus likely induces quicker protective immune response than that induced by a non-related virus such as the one found in the MLV vaccine (Cano et al., 2007a; Cano et al., 2007b).

Whole-herd exposure with attenuated virus vaccine has the advantage of potentially causing fewer clinical signs after exposure and potentially, lower the risk of regional field virus dissemination (Linhares et al., 2012b). Further studies to compare effectiveness of LVI vs. MLV in regards to quantifying the production impact are needed.

During the period of data collection, veterinarians from participating herds reported their perception that “PRRS viruses of RFLP pattern 1-4-4 tended to persist longer in

infected populations compared to those of other RFLP patterns". About half of the enrolled herds were infected with RFLP pattern 1-4-4. However, both univariate and multivariate analyses did not reveal an association between infection with PRRSv RLFP pattern 1-4-4 and TTS. Further studies using more discriminatory molecular techniques to differentiate PRRSv isolates could reveal RNA segments associated with TTS.

Based on an annual herd replacement rate of 40-50%, we assumed that herds that had not detected PRRSv infection for the last 3 years had little to no remaining herd immunity to PRRSv. Thus, we created a dichotomous variable to differentiate herds in regards to "prior PRRSv infection in the past 3 years". The results showed that herds with prior infection achieved TTS before than herds that had not reported prior infection (low immunity herds). Prior infected pigs develop protective immunity to PRRSv faster than pigs infected for the first time (Murtaugh et al., 2002). Therefore, we speculate that a similar phenomenon happens at the herd level. Prior immunity present in herds with prior infection may explain the fact that those herds achieved TTS first due to the development of adaptive immune response present in previously exposed pigs.

Assuming that "prior PRRSv herd immunity" was the reason why herds with "prior PRRSv infection" achieved TTS sooner than herds without prior PRRSv infection (or herd immunity), we highlight the need of further studies to better describe how long PRRSv immunity lasts at a population level and, similarly, what is the minimum period of time necessary for the protective effect of "prior infection status" on TTS. We speculate that factors that affect anti-PRRSv herd immunity level such as PRRSv

immune status of replacement animals and breeding herd replacement rate play an important role on the effect of herd immunity on TTS.

However, it has been reported that prior exposure to pathogens can also result in an epidemic that is longer in duration than if the pathogen was introduced into a naive population (Pulliam et al., 2007). We speculate that this phenomenon called "epidemic enhancement" did not happen in the studied populations because in this study the whole herd was exposed to the virus at once and herds were closed to the introduction of PRRSv-susceptible replacement pigs for a period of time sufficient to significantly decrease the risk of PRRSv-shedding and transmission events within the herd.

"Time to PRRSv-stability" in the pre-weaned pig population is an important parameter to assist producers to make informed decision in regards to which protocol to use to control and eliminate PRRSv from breeding herds. However, other important aspects must be considered when deciding between protocols, including (a) if the protocol has an effect on the clinical impact, and (b) the risk of exposing neighbors to new virus isolates. This study showed the importance of monitoring PRRSv frequently over time in herds undergoing virus elimination. As previously discussed, further studies are needed to explore the variability of TTS among herds.

Strengths of this study include the nature of the investigation which was an applied quasi-experimental study performed with a relatively high number of herds under field conditions, with well defined inclusion criteria.

Confounding variables such as building and equipment design, level of biosecurity, training of staff, PRRSv isolate, and regional density may vary between exposure groups and could lead to misinterpretation of the findings.

Some of the limitations of this study are that (a) monitoring duration and frequency may have missed some infected herds, (b) the existence of unknown confounder variables might have biased the findings, (c) the allocation of treatment (LVI or MLV) was not random and (d) some farms with relatively long TTS might have been infected with an unrelated PRRSv, which was not identified using the monitoring scheme in place. Moreover, participating herds did not submit samples to the same diagnostic laboratory. Therefore, differences in PCR sensitivity might have influenced results (Gerber et al., 2012).

Conclusions

This study provided relevant information on the effectiveness of two different PRRSv immunization approaches on the time it takes to produce PRRS-negative pigs at weaning from acutely infected breeding herds going through PRRSv elimination using the load-close-expose. This information is especially relevant in the context of assisting producers and veterinarians in their regional efforts to eliminate PRRSv from infected areas. Increasing herd closure success rate and decreasing TTS will be pivotal for increasing success of controlling PRRSv within regional programs.

Farms that fulfilled the eligibility criteria of this study achieved TTS sooner if breeding animals had prior contact with PRRSv and when LVI was the method of whole-herd exposure. Moreover, herds assisted by a specific veterinary clinic

achieved TTS sooner than all other herds. Some other factors such as frequency of PRRSv infection, area density of pig farms and pig flow could also influence the decision of veterinarians to determine between immunization treatments in order to control and eliminate PRRSv from infected breeding herds.

Tables and figure

Table 1 – Baseline demographic characteristics of the herds enrolled in the study.

Characteristic	LVI	MLV
Number enrolled	41	20
Prior infection	20 (49%)[a]	13 (65%)[a]
RFLP strain 1-4-4	15 (38%)[a]	12 (60%)[a]
Herd size (Mean ± SE)	3,400 ± 303[a]	2,471 ± 250[b]
Days from PRRSv detection to intervention (Mean ± SE)	28 ± 3[a]	39 ± 9[a]

[a] There were no significant differences between groups at alpha level of 0.05 for prior infection and RFLP 1-4-4 (Fisher's exact, p-values 0.2813 and 0.1102 respectively), and for days from PRRSv-detection to intervention, (t-test, p-value 0.2526). For herd size, LVI herds were significantly larger than MLVs (t-test p-value 0.0324). This sample size allowed a detection of a hazard ratio (to achieve TTS) of at least 2.3 between treatment groups with 80% statistical power.

Table 2 – Number of herds discontinued from the study based on the different reasons by treatment group.

Reason	LVI	MLV
Herds that did not reach TTS by the end of the observational study period	1	3
Detection of a PRRSv with <97% homology compared to the virus identified at the beginning of the LCH program*	3	2
Producer decided to re-start LCH program	2	2
Other		1**

* New virus introduction, as judged by herd veterinarian based on phylogenetic characteristic of PRRSv and clinical signs; ** Pig farm caught on fire and herd had to be depopulated.

Table 3 – Number of herds (and percentage of total herds) by treatment group with intermittent pattern of PRRSv RNA detection by RT-PCR according to detection pattern.

Pattern of PRRSv RNA by RT-PCR detection	LVI	MLV
Straight 4 consecutive negative monthly tests (TTS) (pattern: + -- - -)*	28 (68%)	9 (45%)
One negative monthly test followed by a positive test (pattern: + - +)	13 (32%)	7 (35%)
Two negative monthly tests followed by a positive test (pattern + - - +)	0 (0%)	2 (10%)
Three negative monthly tests followed by a positive test (pattern + - - - +)	0 (0%)	2 (10%)

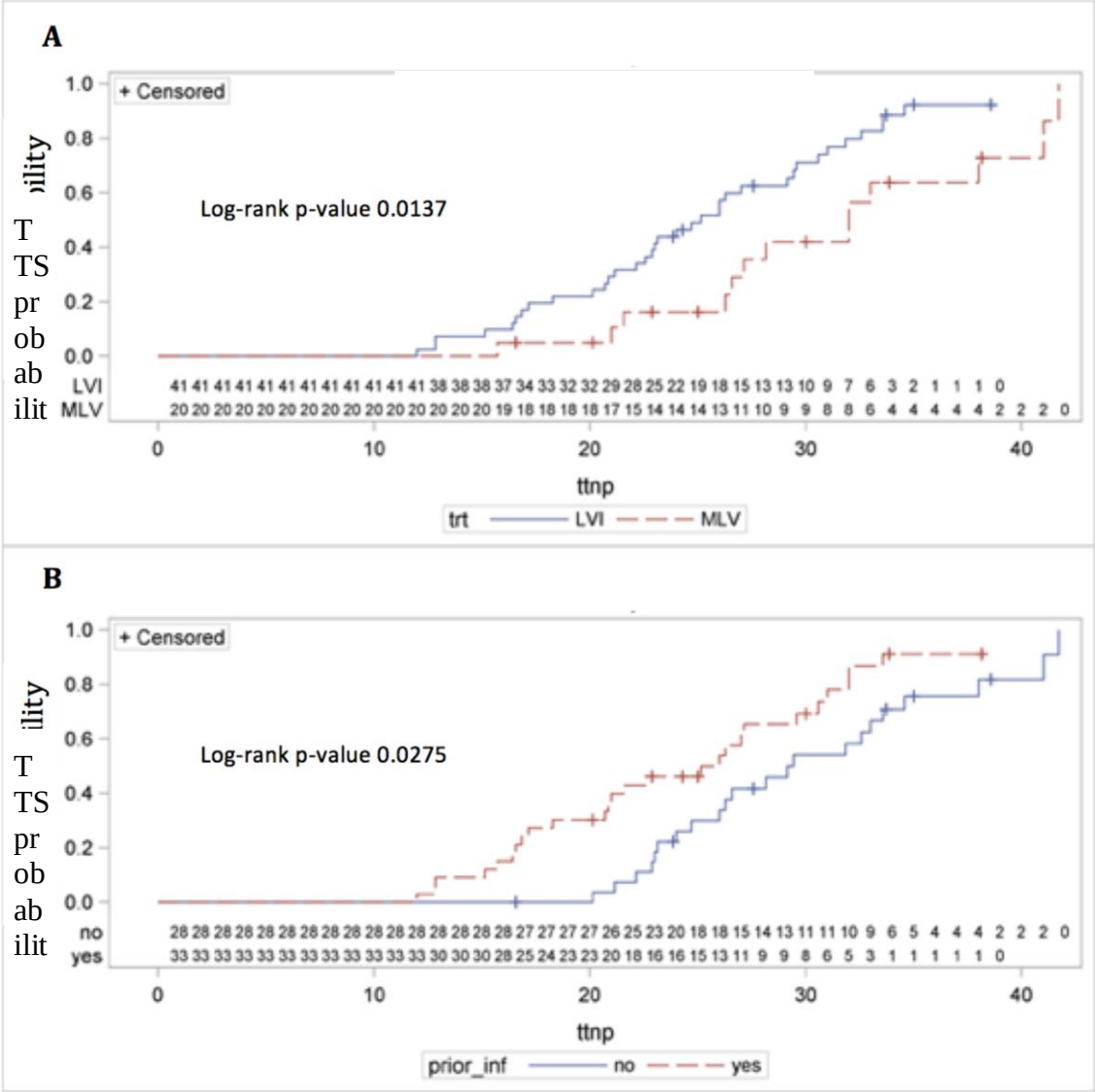
* The pattern in parenthesis represents monthly PRRSv-RNA PCR results: the plus sign (+) represents a month with at least 1 PCR-positive result. Negative sign (-) represents a month with all PCR results being negative for PRRSv-RNA.

Table 4 – Non-significant Kaplan-Meier survival estimates for TTS.

Variable	**Median TTS***	**Log-rank p-value**
RFLP strain 1-4-4 status		
Herds belonging to RFLP 1-4-4	27.0 (21.6, 32.6)	0.5306
Herds with RFLP other than 1-4-4	28.1 (20.1, 38.0)	
Herd size		
Herds < 3000 sows (median size all enrolled herds)	26.6 (21.1, 32.0)	0.4455
Herds ≥ 3000 sows	27.14 (21.85, 34.57)	
Days from infection to intervention (larger or smaller than median interval)		
< 26 days (median interval)	29.4 (21.6, 33.6)	0.5671
≥ 26 days	26.0 (21.0, 32.6)	

* Median TTS and 25th to 75th percentiles.

Figure 1 - Kaplan-Meier estimates for time-to-PRRSv-stability (TTS) according to whole-herd exposure treatment (A) and herd status prior to PRRSv-infection (B).



The log-rank p-values included in each plot tested the whole survival curves and mean that TTS was significantly different according to a) treatment type or b) prior PRRSv infection status. Panel A: Median TTS and interquartile range (25th to 75th percentiles) for herds treated with LVI and MLV were 25.1 weeks (20.7, 31.0) and 32.0 weeks (26.6, 41.0), respectively. Panel B: Median TTS and interquartile range for herds with and without prior PRRSv-infection history were 26.0 weeks (17.1 to 31.0 weeks) and 29.3 weeks (24.0 to 34.6 weeks), respectively.

Chapter III - Comparison of production losses between whole load-close-expose programs to control PRRSv in breeding herds

Linhares DCL, Cano JP, Torremorell M, Morrison RB. Comparison of production losses between whole herd exposure programs to control PRRSv. [in prep, to be submitted to Prev Vet Med].

Abstract

Porcine reproductive and respiratory syndrome virus (PRRSv) is a pathogen associated with significant production losses. To control and eliminate PRRSv from breeding herds, some veterinarians have adopted a strategy called load-close-expose which consists of interrupting replacement pig introduction in the herd for several months and exposing the whole herd to a replicating PRRSv. This was a prospective study that followed breeding herds acutely infected with PRRSv that adopted load-close-expose with either a modified live virus (MLV) vaccine (n=18) or live-resident virus inoculation (LVI) (n=40). The treatment groups (load-close-expose with MLV or LVI) were compared for: a) the time to baseline production (TTBP), which was defined using exponentially weighted moving average method and represented time to recover to the number of pigs weaned per week that herds had prior to PRRSv-detection, and b) the total production loss in terms of number of pigs weaned per week. TTBP was compared between treatments using Kaplan-Meir survival curves and log-rank test. A multivariate analysis using proportional hazards regression was performed adjusting the effect of treatment on TTBP to 'severity of PRRSv infection', 'number of whole-herd exposures', 'days from PRRSv-detection to intervention', 'prior PRRSv-infection status' and 'veterinary clinic responsible for technical assistance for the herd'. Total loss was compared between groups using multivariate regression analysis adjusted by the covariates considered for the TTBP analysis. The overall TTBP was 16.5 weeks (range 0 to 29 weeks). The magnitude of production losses following whole-herd exposure averaged 2,217 pigs not weaned/1,000 sows and had a high correlation with TTBP. Herds in the MLV group recovered production sooner and had less total loss than herds in the LVI group.

Similarly, TTBP was significantly sooner and the total loss significantly less in herds assisted by a specific veterinary clinic and herds that were infected with PRRSv in the 3 years prior to the study. This study provided new metrics to assist veterinarians to decide between methods of load-close expose to control and eliminate PRRSv from breeding herds. For that, it developed new parameters and methods that can be used in future studies to compare effect of interventions on livestock productivity.

Keywords: PRRSv, control, elimination, load-close-expose, whole herd exposure, live virus inoculation, serum inoculation, modified-live virus vaccine, SPC, EWMA.

Introduction

The annual cost of porcine reproductive and respiratory syndrome virus (PRRSv) to the US swine industry was estimated at 664 million dollars (NPB, 2011). To reduce production impact, different strategies to control and eliminate PRRSv at herd and regional levels have been employed (Corzo et al., 2010; Rowland and Morrison, 2012). In North America, herd closure combined with whole-herd exposure to a live PRRSv is a commonly attempted strategy to control and eliminate the virus from infected breeding herds (Yeske, 2009).

The concept of herd closure is straightforward: pig introduction in breeding herds is interrupted for a period of time to allow infection to die out. The underlying principle is to prevent the introduction of susceptible pigs that, in contact with PRRSv, could get infected and disseminate virus within the herd perpetuating the infection (Torremorell et al., 2003). The strategy of whole-herd exposure was developed to be an addition to the traditional herd closure approach and has the principle of exposing all pigs in the herd with a live PRRSv immunogen (Desrosiers and Boutin, 2002). As a consequence, it is expected that most pigs will get infected, become infectious in the short term and immune in the long term; therefore, homogenizing the infection and immunity statuses of animals in the population. When herd closure is combined with whole-herd exposure (strategy called load-close-expose), the expectation is that protective immunity is built at the population level resulting in the decrease and eventual elimination of PRRSv from the breeding herd. Consequently, newborn piglets do not become infected and thus are weaned “PRRSv negative” (Holtkamp et al., 2011).

Two immunogens commonly used for PRRSv whole-herd exposure in North America's swine industry are PRRSv modified-live virus (MLV) vaccine and the live-resident herd virus. Linhares et al reported that herds that used live-resident virus inoculation (LVI) in load-close-expose programs to eliminate PRRSv achieved PRRSv-stability status sooner than herds that used MLV protocols (Linhares et al., 2012a). However, it is still unknown which protocol leads to a lesser production impact in the breeding herd. The objectives of this study were to compare: (a) the time it took for farms to recover the production levels they had prior to PRRSv infection (time to baseline production, TTBP) and (b) the magnitude of the production losses following PRRSv infection and adoption of load-close-expose strategy using LVI or MLV. Productivity was measured as "total number of pigs weaned per 1,000 sows" per week (Schaefer and Morrison, 2007).

To the best of our knowledge, this is the first study to use statistical process control methods to define return to "in control" levels of livestock production parameters. The methods described in this manuscript can be used in future studies to evaluate effectiveness of different preventive or therapeutic strategies on production impact of diseases.

Design and methods

Study design and source population

This was a prospective, quasi-experimental design with 58 swine breeding herds enrolled to compare changes in number of pigs weaned following PRRSv-infection and use of load-close-expose program as part of PRRSv elimination programs.

Change in ‘number of pigs weaned per 1,000 sows per week’ was compared between herds that used MLV vaccine (n=18) to that of herds that used LVI (n=40) (table 1).

More specifically, we compared MLV and LVI exposure programs for: a) the time it took to recover the number of pigs weaned per week prior to PRRSv-detection, and b) the total production losses in terms of number of pigs weaned. To perform the analysis, 3 parameters were developed: time to baseline production (TTBP), delta pigs A (Δ pigs A) which represented severity of PRRSv infection and delta pigs B (Δ pigs B) which was a measure of total production loss from load-close-expose to TTBP (table 2).

Veterinarians listed in the American Association of Swine Veterinarians (AASV) directory were invited via email to enroll herds to which they provided technical services. Eligible herds were those acutely infected with PRRSv from November 2009 to March 2012 and that adopted load-close-expose program with intent to produce PRRSv-negative pigs at weaning. Enrolled herds fulfilled the following inclusion criteria: a) stable sow inventory (herd not expanding or shrinking by more than 10 %); b) diagnostic evidence that PRRSv was present; c) intent of herd owner to eliminate PRRSv from the herd; d) no replacement pigs introduced during the closure period; e) owners willing to share production performance data and f) enrollment in the study within 2 months of starting the load-close-expose program.

The study was approved by the University of Minnesota’s IACUC and all the farm owners or their representatives provided written, informed consent agreeing to participate.

PRRSv used for breeding herd exposure

Herds in the MLV group were vaccinated with either Ingelvac® PRRS ATP (Boehringer Ingelheim Vetmedica Inc., St. Joseph, MO) vaccine or Ingelvac® PRRS MLV (Boehringer Ingelheim Vetmedica Inc., St. Joseph, MO) vaccine. LVI protocols were based on application of diluted PRRSv collected from serum from infected pigs from the farm (Batista et al., 2002).

Statistical analysis

Exponentially weighted moving average (EWMA) charts were constructed to define TTBP with the intention of operating with a low false-alarm rate (i.e. detect 'back to *in control'* with high specificity) and thus using a weight (λ) of 0.40 and a multiple of sigma (σ) of 3 for control limits (Krieter et al., 2009). Such parameters resulted in an average time to signal of 1.97 weeks, meaning that in the long run, the control chart required no more than approximately 2 weeks to detect TTBP.

TTBP was compared between groups with the use of Kaplan-Meier survival estimates. Differences in survival curves between groups were assessed with the use of log-rank test. Moreover, Cox proportional-hazards regression was used to calculate the magnitude of the treatment effect on TTBP and to adjust analysis for covariates. The following covariates were considered in the analysis: 'Δ pigs A' (delta pigs A), 'number of whole-herd exposures', 'days from PRRSv-detection to intervention', 'prior infection status' and 'veterinary clinic responsible for technical assistance for the herd'. Factors associated with TTBP with a p-value less than 0.20 in the

univariate analysis were entered in the multivariate model and non-significant factors were removed by means of a backward-selection procedure.

The outcome 'Δ pigs B' (delta pigs B) was compared between treatment groups using t-test. General linear models were constructed to adjust the effect of treatment on 'Δ pigs B' by the same covariates considered in the TTBP analysis.

Herds that were PRRSv-free (AASV category IV) (Holtkamp et al., 2011) during the previous 3 years prior to the present study were considered to have "no prior PRRSv infection". Herds with at least one PRRSv introduction in the past 3 years before the study were considered to have "PRRSv prior infection".

Spearman correlations were calculated between TTBP, "Δ pigs A", "Δ pigs B" and "days from PRRSv-detection to load-close-expose". Since this study was conducted with the same herds enrolled in a previous study that calculated time-to-PRRSv-stability (TTS), it was possible to correlate data from this study to TTS of the same herds (Linhares et al., 2012a). Briefly, TTS was defined as number of weeks it took to achieve failure to detect PRRSv RNA in pre-weaning pigs by RT-PCR for a 90-day period by collecting 30 monthly serum samples, considering day 1 the day that the herd was exposed to either MLV or LVI.

Two-sided hypotheses and tests were used for all statistical inferences. A p-value of less than 0.05 was considered to indicate statistical significance. Statistical analyses were performed using the SAS 9.2 package (SAS Institute, Inc., Cary, NC).

Results

Descriptive analysis

The overall median TTBP was 16.5 weeks and ranged from 0 to 29 weeks. The magnitude of production losses following whole-herd exposure (Δ pigs B) ranged from 0.0 to 9,561.0 pigs/1,000 inventoried sows and averaged 2,217.2 pigs not weaned/1,000 sows.

The average number of pigs not weaned from PRRSv detection to exposure (Δ pigs A) was 571.9 pigs/1,000 sows. The Δ pigs A parameter was slightly higher in the LVI group ($p = 0.0681$) (Table 3).

TTBP had a Spearman correlation coefficient of 0.406 (p 0.0016) and 0.885 ($p < 0.0001$) with Δ pigs A and Δ pigs B, respectively. The correlation between TTBP and TTS was 0.594 ($p = 0.0002$) for herds in the LVI group and 0.0850 ($p = 0.7557$) in the MLV group. The correlation of TTBP and time from PRRSv-detection to treatment was -0.201 ($p = 0.1305$).

Univariate analysis

Herds in the MLV group recovered production sooner ($p < 0.0001$) and had less total loss ($p = 0.0095$) than herds in the LVI group (Table 3). Similarly, herds assisted by veterinarians from a specific veterinary clinic reached TTBP sooner and had less production losses due to PRRSv than herds assisted by other veterinarians ($p = 0.0031$ and $p = 0.0109$ respectively). Herds 'with' and 'without' prior PRRSv infection had a median TTBP of 11 weeks (25th to 75th percentile: 5 to 20) and 21 weeks (25th to 75th percentile: 16 to 24) ($p = 0.0014$), respectively. Herds with prior PRRSv infection had 1,907 pigs/1,000 sows less total loss than herds without prior PRRSv infection ($p = 0.0001$).

Multivariate analysis

The proportional hazards regression model indicated that all variables significant in the univariate analysis maintained significance after being adjusted by each other (Table 4).

Similarly, a multivariate general linear model that adjusted the effect of Δ pigs A, treatment, prior infection status and veterinary clinic resulted in all the parameters remaining significant predictors of Δ pigs B after adjustment by each other (p-values = 0.0114, 0.0285, 0.0132 and 0.0040 respectively).

Discussion

This study was part of a prospective study where the effectiveness of "PRRSv herd closure" programs was evaluated. Breeding herds acutely infected with PRRSv that adopted load-close-expose programs with either LVI or MLV with intent to produce PRRSv negative pigs at weaning were followed. In a previous study (phase 1), we compared effectiveness of load-close-expose programs in terms of time to reach PRRSv-stability (TTS). We reported that TTS was shorter in herds that: a) were treated with LVI, b) had history of at least one PRRSv infection in the previous 3 years from the study and c) were assisted by a specific veterinary clinic (Linhares et al., 2012a). In the second phase (present study), we compared the impact of the exposure method on the time it took to achieve baseline production (TTBP) and on the magnitude of the production losses associated with PRRSv infection. Outcomes of this project will help producers to make informed decisions on which load-close-expose strategy to use to control and eliminate PRRSv from breeding herds.

The correlation coefficients between TTBP and Δ pigs A and Δ pigs B were weak and strong, respectively (Taylor, 1990) meaning that the severity of the initial production loss could not predict TTBP. However, the higher the total loss following load-close-expose, the longer it took for herds to recover productivity.

The correlation between TTBP and TTS depended on the treatment being weak for MLV and moderate for LVI (Taylor, 1990). This suggests that productivity of weaned pigs was a poor indicator of the PRRSv status of the herd.

Some herds showed no statistically significant change in productivity following PRRSv infection and adoption of load-close-expose programs, while other herds required a relatively long time to recover production levels (reach TTBP) and had severe production losses (as high as 9,561 pigs/1,000 sows). Our study differed from a previous report where PRRSv elimination from a breeding herd using load-close-expose with MLV vaccine had no impact on number of pigs weaned (Voglmayr et al., 2006).

A study done by Schaefer and Morrison comparing total number of pigs weaned 52 weeks before and 52 weeks after herd closure implementation to eliminate PRRSv from infected breeding herds found that 13 out of 15 herds weaned similar or more pigs after the herd closure program (Schaefer and Morrison, 2007). The main differences between that study and the present one are that here breeding herds were acutely infected with PRRSv and adopted load-close-expose whereas in Schaefer and Morrison's study herds were endemically infected and used herd closure only (no deliberated exposure to live PRRSv).

Herds in the MLV group had numerically smaller Δ pigs A (delta pigs A) than LVI herds indicating that severity of PRRSv infection at the moment of load-close-expose was less for MLV herds. Severity of PRRSv break might be one of the factors that influenced veterinarian's decision to choose between MLV or LVI as immunogen for load-close-expose programs. Therefore, Δ pigs A was included as a covariate in the multivariate regression models to compare TTBP and Δ pigs B between treatment groups, adjusting effects to that variable.

Our previous study showed that herds in the MLV program had at least one PRRSv RNA RT-PCR negative test followed by positive result (Chapter 2). This might be explained by the fact that MLV herds recovered productivity sooner than LVI herds, suggesting that PRRSv infection prevalence was lower in the MLV group compared to LVI group.

We speculate that because production levels recovered sooner than TTS in MLV herds, farm personnel might have assumed that PRRSv was already eliminated from the herd. Thus, extra management practices adopted at the herd to limit within farm transmission of PRRSv (such as cross fostering of piglets, washing and down time of farrowing crates) were relaxed sooner than achievement of PRRSv negative status, favoring re-circulation of PRRSv and contributing to a longer TTS for MLV herds.

The fact that herds assisted by a specific veterinary clinic reached TTBP sooner and had a less severe total loss might be explained by specific standard operating procedures followed by such herds during herd closure. Therefore there is a need to study the relationship between management practices and herd closure success.

Statistical process control (SPC) charts have been used in veterinary medicine to monitor changes in key performance indicators related to livestock production (De Vries and Reneau, 2010). There are different types of SPC charts and the most commonly described are Shewhart and EWMA (De Vries and Reneau, 2010). The advantages of EWMA over Shewhart include that EWMA has fewer false alarms, is more flexible allowing adjustments for better sensitivity or better specificity, and there is no need for extra sensitizing rules (Montgomery, 2012). Also, when EWMA is used for individual measurements, which was the case in this study (number of pigs weaned per week), it is considered a distribution free (nonparametric) procedure (Montgomery, 2012). In the swine industry, EWMA has been used to identify small deviations in repeat estrus rate and total born per litter (Krieter et al., 2009). However, this is the first report of a SPC tool to detect "back to control" state of a production parameter.

A limitation of this study is that it was assumed that PRRSv was the primary cause of production losses compared to the reference period. Thus, herds were not monitored to other porcine pathogens. Moreover, to calculate TTBP the baseline levels were arbitrarily set to the average of 21 weeks prior to PRRSv-detection. Analysis with herds from this study that were PRRSv-negative (AASV category IV) (Holtkamp et al., 2011) showed a high correlation between 52 and 21 week-average in the total number of weaned pigs (data not shown). We chose 21 weeks because it represents a full breeding cycle and because not all herds had stable sow inventory during 52 weeks prior to the study.

This study provided important metrics on load-close-expose projects using LVI or MLV as methods of whole-herd exposure. Time it takes to reach PRRSv stable herd (Linhares et al., 2012a), time it takes to recover production levels that the herds had before PRRSv infection and number of pigs “not” weaned were described. Further studies are needed to elucidate the effect of management practices on herd closure success as well as to develop economic models using information here described to assist in the decision between using LVI or MLV as part of load-close-expose programs to control and eliminate PRRSv from breeding herds.

Conclusions

Herds that used MLV vaccine as part of PRRSv control programs had less severe production losses following PRRSv infection and load-close-expose programs compared to farms that used LVI. Similarly, TTBP was significantly shorter and productivity was less affected in farms with prior PRRSv infection and herds assisted by a specific veterinary clinic. Moreover, the correlation between TTBP and TTS was significant for herds using LVI but not significant for herds in the MLV program, as herds in the MLV had a variable and often longer period from TTBP to TTS status compared to LVI group. Further studies are needed to better understand the association of management practices with herd closure success.

Tables

Table 1 – Baseline demographic characteristics of enrolled herds before establishment of load-close-expose as part of PRRSv elimination programs

Characteristic	LVI	MLV
Number enrolled	40	18
Prior infection	20 (50%)	12 (65%)
RFLP strain 1-4-4	15 (38%)	10 (55%)
Herd size (Mean ± SE)	3,400 ± 303	2,471 ± 250*
Days from PRRSv detection to intervention (Mean ± SE)	28 ± 3	39 ± 9

* Difference between groups was statistically different at alpha level of 0.05.

Table 2. Parameters used to characterize change in total pigs weaned per week.

Parameter	Meaning	Method of calculation
Time to baseline production (TTBP)	Time, in weeks, it took to recover the levels of 'weaned pigs per week' that the herd had prior to PRRSv-detection (i.e. time to "in control" levels of productivity*).	Defined by statistical process control using the exponentially weighted moving average (EWMA) control chart method (Montgomery, 2012)**.
"Δ pigs A"	Total number of pigs not weaned/1,000 sows from the time of PRRSv-detection to time of herd exposure to LVI or MLV. It indicated the severity of PRRSv break and was used to adjust comparisons of the "Δ pigs B" parameter.	Calculated by adding the sums of total number of pigs weaned below the expected value (21 week-average before PRRSv-infection) from time of PRRSv-detection until whole-herd exposure to either LVI or MLV.
"Δ pigs B"	Total number of pigs not weaned/1,000 sows from the time of whole-herd exposure to TTBP status achievement. It represented the 'total production loss' following PRRSv infection and load-close-expose using LVI or MLV.	Cumulative sum of number of pigs weaned below the expected from the time of whole-herd exposure to TTBP status.

* For the purpose of this study, productivity was measured as quantity of pigs weaned per week. ** Details on the parameters used to construct EWMA are provided in the 'statistical analysis' subsection of the materials and methods section.

Table 3 – Comparison of PRRSv severity (Δ pigs A), total loss (Δ pigs B) and time to baseline production (TTBP) between groups (univariate analysis).

	LVI	MLV	p-value*
Δ pigs A (mean ± std. error)	678.4 ± 106.0	335.3 ± 141.4	0.0681
Δ pigs B (mean ± std. error)	2665.0 ± 313.0	1222.2 ± 395.3	0.0095
TTBP (median and 25th to 75th percentile)	21 (13, 24)	10 (0, 15)	<0.0001

* Between group comparisons for Δ pigs A and Δ pigs B were done using t-test and for TTBP using Log-rank test.

Table 4 – Adjusted effects of treatment and other covariates on TTBP (proportional hazards regression analysis).

Parameter	Hazard ratio	p-value*
Δ pigs A	0.99	0.1000
Treatment (LVI vs. MLV)	3.59	0.0005
Prior infection status (yes vs. no)	2.50	0.0031
Veterinary clinic	3.39	0.0007

* Chi-square p-values from Type III tests of Cox proportional hazards regression.

Chapter III Addendum - Economic analysis of attenuated PRRS virus vaccine compared to live resident virus inoculation as part of a load-close-expose program to eliminate PRRSv from breeding herds.

Linhares DCL., Morrison RB. Economic analysis of attenuated PRRSv vaccine compared to live-resident virus inoculation as part of a load-close-expose program to eliminate PRRSv from breeding herds. [in prep, JSHAP].

Introduction

Porcine reproductive and respiratory syndrome virus (PRRSv) is a pathogen associated with significant production losses, which results in substantial increases in production costs (Neumann et al., 2005). To control and eliminate PRRSv from breeding herds, some veterinarians have adopted a strategy called load-close-expose which consists of interrupting replacement pig introduction in the herd for several months (herd closure) and exposing the whole herd to a replicating PRRSv. Either modified-live virus (MLV) vaccine or live-virus inoculation (LVI) is used (Corzo et al., 2010; Desrosiers and Boutin, 2002; Torremorell et al., 2003).

Herds that used MLV required 7 additional weeks to reach PRRSv-stability compared to herds that used LVI (Chapter 2). However, MLV herds recovered production levels 11 weeks sooner and had less total loss in pigs weaned (advantage of 1,443 pigs per 1,000 sows) (Chapter 3). There is the need to determine which program (MLV or LVI) had better overall economic advantage.

Partial budgeting is a method commonly used in veterinary medicine to determine the economic benefit of interventions (Chaminuka et al., 2012; Li et al., 2012; Rodgers et al., 2012; Sami et al., 2012). Basically, partial budget analysis takes into account: a) increase in income, b) reduction of elimination of income, c) reduction or elimination of costs and, d) increase in costs (Abbitt and Townsend, 1979).

This study consisted of a partial budget analysis to compare MLV to LVI as the whole herd exposure method of a load-close-expose program to control and eliminate PRRSv from infected breeding herds.

Methods

Modeling approach.

A partial budget analysis was applied to conduct deterministic simulations (model A) based on net margin over feed costs for a breeding herd of 1,000 sows if MLV is chosen vs. LVI as exposure method of a load-close-expose program. A stochastic model using Monte Carlo simulation was also conducted (model B).

Model A.

Input variables.

Outcome variables from the literature and from Chapters 2 and 3 of this thesis were used as input variables for this economic model (Table 1).

Calculations.

MLV vaccine was estimated to cost $1.00 per dose with 3 doses being administered per sow. For LVI, $100 was assumed for extra diagnostic costs to test the PRRSv inoculum used for herd exposure.

The opportunity cost for pigs not weaned was estimated by margin over cost of feed not consumed. Market price of $80.00/cwt carcass, average carcass weight of 200 lbs, and wean to finish feed cost of $ 99.49 per pig were assumed to yield an opportunity cost of $66.72 per pig not weaned.

The opportunity cost for impeded growing performance was calculated by multiplying the number of weeks it took for herds to achieve "TTS" status[2] by the "number of pigs weaned per week" (500) and "cost attributed to PRRSv infection" in growing pigs ($13.52) (Neumann et al., 2005). To avoid double counting, "pigs not weaned" from "TTBP"[3] to "TTS" was subtracted from the calculation, resulting in the formula listed below:

Opportunity cost for impeded growing performance	=	((TTS – TTBP) * pigs weaned per week * $13.52) + ((TTBP * pigs weaned per week – pigs not weaned per 1,000 sows) * $13.52)

The overall opportunity cost was obtained by adding the opportunity costs for pigs not weaned and for reduced growing performance. Then, the opportunity cost for farms that were treated with LVI was subtracted from that of farms that used MLV to show the advantage (or disadvantage) to use MLV instead of LVI.

To describe the relative importance of key variables of the model, sensitivity analyses were performed. The first sensitivity analysis considered differences in the 'extra production cost' attributable to PRRSv-infection and pig margin over variable cost

[2] Herds achieved TTS (time to PRRSv stability), status when obtained 4 consecutive negative PCR results for PRRSv RNA in a monthly basis testing of at least 30 piglet blood samples.
[3] Time to baseline production (TTBP) was defined as time, in weeks, it took to recover the levels of 'weaned pigs per week' that the herd had prior to PRRSv-detection (i.e. time to "in control" levels of productivity).

(MOVC). The second sensitivity analysis looked at weeks for TTS for MLV herds that allowed the economic model to converge at a break-even point with LVI herds.

Model B.

Input variables and calculations.

The same input variables used in the model A were used in model B. However, the model B took into account variability in the TTS, TTBP and total loss parameters (Table 3) using Monte Carlo simulation approach (Metropolis and Ulam, 1949). Kolmogorov-Smirnov test on data from previous chapters indicated that the TTS, TTBP and total loss parameters fitted Normal distributions at alpha level of 0.05, and therefore, the distributions assumed for these 3 parameters was Normal. The calculations of opportunity costs were performed using the same formulas described in model A. However, because model B is stochastic, the outcome was a distribution that took 3,000 random samples (iterations) from the TTS, TTBP and total loss distributions. To execute the model B, the RiskSim ® v 2.43 software (TreePlan Inc, San Francisco, CA) was used.

Results and discussion

Model A showed that the longer TTS observed in herds exposed with MLV compared to LVI herds resulted in approximately $ 67,000 disadvantage per 1,000 sows. Conversely, the lower total production loss in MLV herds compared to LVI herds resulted in approximately $ 96,000 per 1,000 sows advantage. The net advantage for MLV herds was $ 26,548 per 1,000 sows (Table 2). These two parameters influenced the break-even between LVI and MLV programs in opposite directions: rising

PRRSv-attributable cost and rising MOVC reduces and increases the advantage of MLV, respectively (Figure 1).

Cost attributable to PRRSv represents extra production costs that PRRSv infected pigs have due to higher mortality, higher feed conversion rates and slower growing performance. The major contributor to variable costs in pig production is feed (USDA, 2004). Besides feeding, health interventions and trucking also contribute to MOVC.

MLV holds economic advantage over LVI when TTS is below 38 weeks (Figure 2).

Model A was a simple, deterministic model and did not take into account covariates such as prior infection status for PRRSv or days from infection to intervention.

The stochastic model (model B) on the other hand, took into account variability on the distributions of TTS, TTBP and total loss (Figure 3). It resulted in economic benefit for MLV on 59.99% of the 3,000 iterations.

Limitations of both models (A and B) lie on the assumptions made for the analysis: it was assumed that herds that reached TTS would have truly eliminated PRRSv and remained PRRSv-free. It was also assumed weekly all in – all out operation of growing pig sites. Moreover, it was considered that in MLV herds, there was virulent PRRSv circulating in the herd until TTS was reached.

These partial budget models took advantage of information generated in the previous two chapters of this thesis and translated our major findings to a economic models to assists veterinarians to choose between MLV and LVI as methods of whole herd exposure in a load-close-expose program to control and eliminate PRRSv from

breeding herds. The models showed that having a less severe production loss (MLV program) was financially more advantageous than reaching PRRSv-negative status sooner (LVI program). Sensitivity analysis showed the relative importance of margin over variable costs (MOVC) and cost per PRRSv-positive pig on the economic analysis between LVI and MLV.

Tables and figures

Table 1 – Input variables for the partial budget model that compared economic outcomes of MLV herds compared to LVI herds in load-close-expose programs to produce PRRSv-negative pigs from infected breeding herds.

Variable	Meaning in the model	Values and source
Time to PRRSv stability (TTS)	As TTS increases, the number of batches of PRRSv-positive pigs weaned increases, impacting the cost of growing pig production	LVI and MLV median TTS from Chapter II, 25 and 32 weeks respectively. Variability was added in the model considering different TTS for MLV herds.
Time to baseline production (TTBP)	Used to adjust number of pigs weaned before TTS was reached	LVI and MLV median TTBP from Chapter III, 10 and 21 weeks respectively.
Total loss (pigs not weaned/1,000 sows)	Pigs not weaned represent decreased revenue	LVI and MLV average total loss from Chapter III, 1,222 and 2,665 pigs respectively.
Cost of PRRSv-positive pig at weaning	PRRSv-positive pigs have extra cost of production	$ 13.52 (Neumann et al., 2005). Also, variability was added in the model considering the following values $ 10, 15 and 20.
Margin over variable cost (MOVC)	Pigs not weaned represent opportunity margin	$66.72 (Morrison, personal information 2012)
Number of pigs weaned per sow per year	Used to calculate productivity level of a sow farm	26.00 pigs weaned/sow/year

* Briefly, it was considered 12 months average of future prices of feed, plus extra variable costs including veterinary services ($4/pig) and trucking ($2/pig).

Table 2 – Partial budget model to compare economic benefit of MLV over LVI on a load-close-expose program to eliminate PRRSv from an infected 1,000 sows breeding herd.

Treatment	Cost to expose		Opportunity Cost for pigs not weaned				Opportunity Cost for w-f performance			
	Exposure		TTBP	Total loss	OC*		TTS	OC		Total OC
MLV	$3,000	+	12	1,222	$81,532	+	32	$199,799	=	$284,330
LVI	$100	+	20	2,665	$177,809	+	25	$132,969	=	$310,878
Difference (MLV-LVI)					**$ 96,277**			**$-66,829**		**$26,548**

* OC = Opportunity cost

Table 3 – Input variables for the stochastic simulation (model B).

Variable	**Distribution**	**Distribution parameters**
Time to PRRSv stability (TTS)	Normal	LVI herds: mean 25.02, Std. Dev 6.91 MLV herds: mean 31.87, Std. Dev 8.84
Time to baseline production (TTBP)	Normal	LVI herds: mean 18.28, Std. Dev 7.56 MLV herds: mean 8.48, Std. Dev 7.18.
Total loss (pigs not weaned/1,000 sows)	Normal	LVI herds: mean 2,665, Std. Dev 1,980 MLV herds: mean 1,222 Std. Dev 1,677
Cost of PRRSv-positive pig at weaning	Fixed (no distribution)	$ 13.52 (Neumann et al., 2005). Also, variability was added in the model considering the following values $ 10, 15 and 20.
Margin over variable cost (MOVC)	Fixed (no distribution)	$66.72 (Morrison, personal information 2012)
Number of pigs weaned per sows per year	Fixed (no distribution)	26.00 pigs weaned/sow/year

Figure 1 – Break-even analysis between LVI and MLV varying margin over variable cost per pig (x-axis) and PRRSv-cost of growing pig (different intercepts).

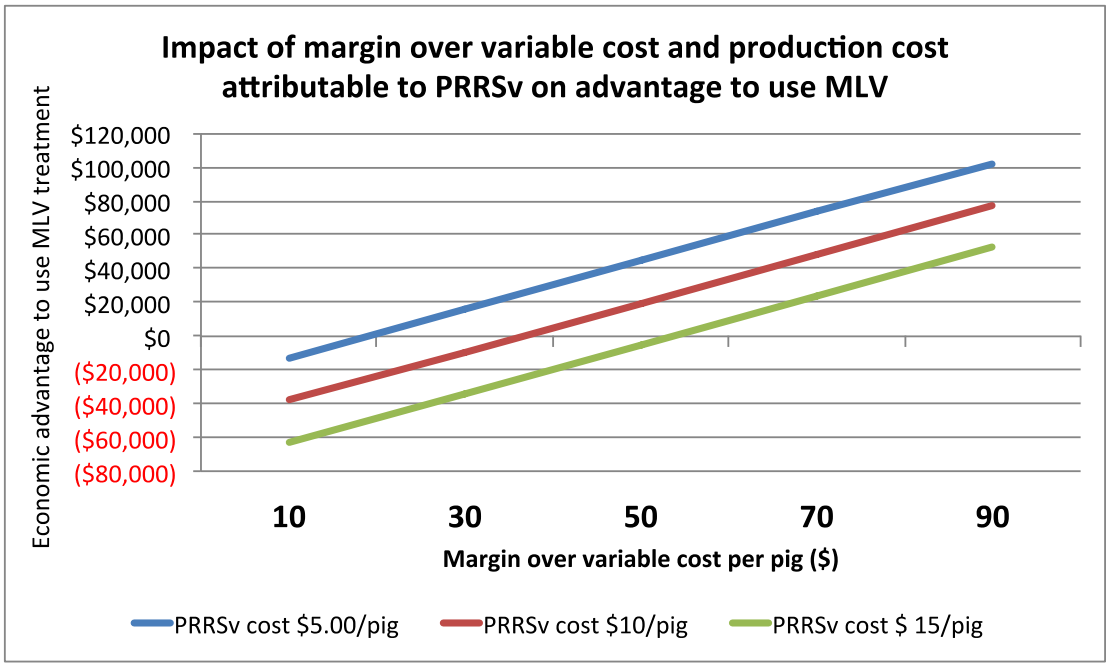


Figure 2 – Break-even analysis between LVI and MLV varying TTS for MLV.

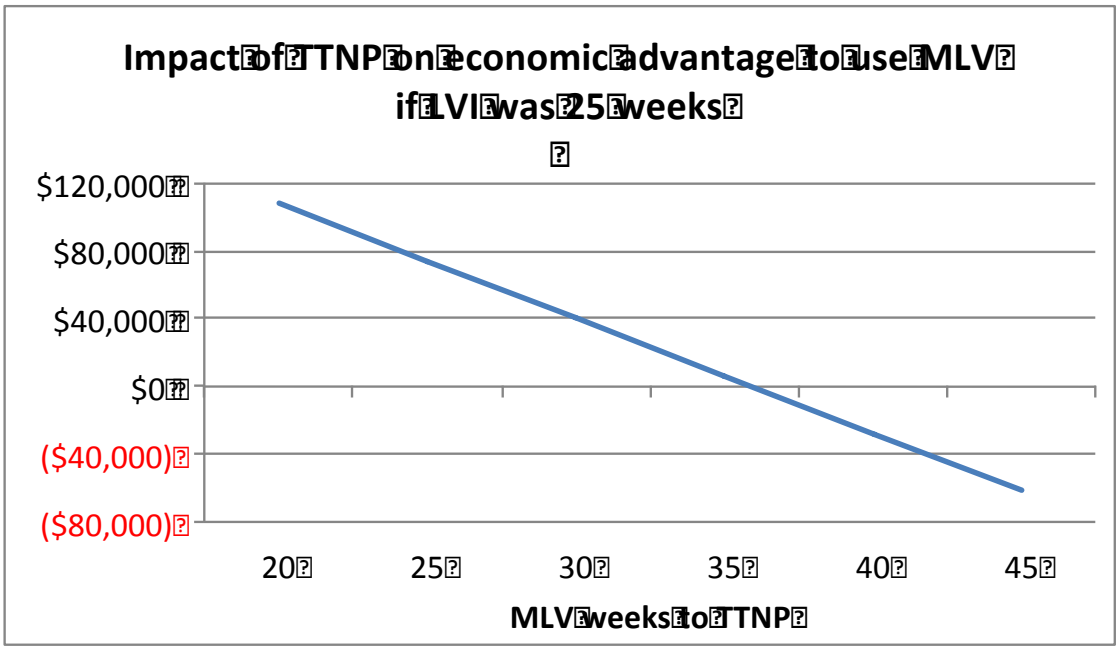
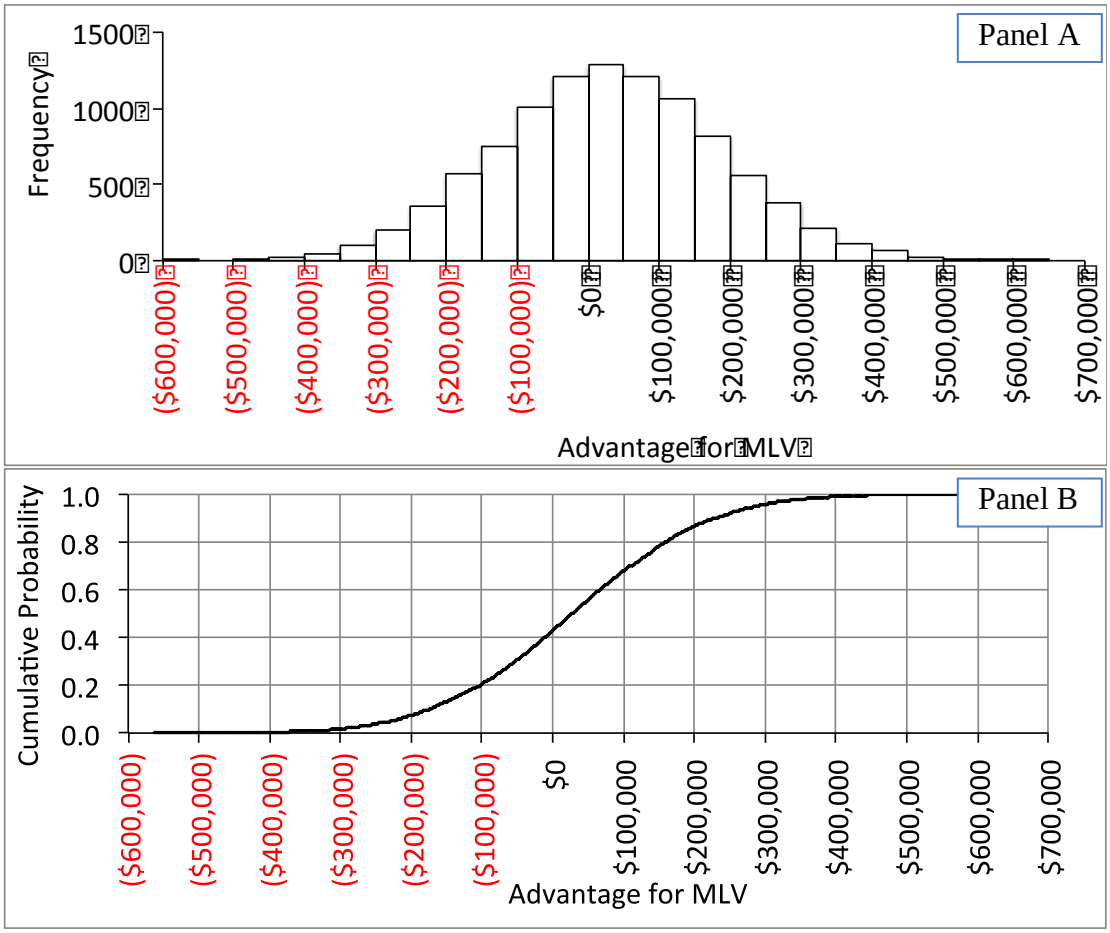


Figure 3 – Distribution of economic advantage of farms that used MLV in comparison to those that used LVI as part of a load-close-expose program*.



* Outcome of model B, a Monte Carlo simulation of economic advantage of MLV compared to LVI. Panel A is the outcome illustrated as a probability density function and Panel B is the outcome illustrated as a cumulative density function.

Chapter IV - Herd closure practices associated with successful PRRSv-elimination from breeding herds

Linhares DCL, Torremorell M, Morrison RB. Herd closure practices associated with successful PRRSv-elimination from breeding herds [in prep].

Abstract

Porcine reproductive and respiratory syndrome virus (PRRSv) is an economically important swine pathogen. Load-close-expose (LCE) is a widely practiced method in North America to control and eliminate PRRSv from breeding herds and consists of interrupting replacement pig introduction for several weeks and exposing the whole herd to either a live-resident virus or a modified live virus (MLV) vaccine.

This study was the phase 3 of a PRRSv herd closure project where breeding herds acutely infected with PRRSv were prospectively followed from Fall 2009 to Winter 2012. The objectives of this study were to a) determine the proportion of sow herds that were still infected with PRRSv despite having failed to detect the virus in weaned pigs for at least 90 days in herds that adopted LCE and b) to evaluate association of management practices recommended by veterinarians with success of LCE programs. For the purpose of this study, "LCE success" was defined as ability of a PRRSv-infected breeding herd that adopted a LCE program to reach the American Association of Swine Veterinarian's category III (PRRSv provisional negative status). Information for this study was obtained from a survey sent to veterinarians overseeing the participating breeding herds.

Forty herds (71 %) that started LCE program with a goal to eliminate PRRSv reached AASV category III. There was no significant difference between herds that used live-resident virus inoculation (LVI) (70%) compared to those that used MLV (75%). Excluding failures associated with detection of new PRRSv, 28 (85%) and 12 (100%) farms using LVI and MLV that reached 90 days of failure to detect PRRSv in pre-weaning piglets after adopting LCE subsequently achieved category III, respectively.

The variables associated with failure to reach AASV category III were a) being infected with a PRRSv of RFLP pattern 1-4-4 and holding back pigs at weaning for quality.

The present study reports the success rate of LCE programs to eliminate PRRSv from breeding herds and highlights the importance of further studies to elucidate important knowledge gaps on PRRSv infection dynamics in pig populations undergoing PRRSv elimination programs.

Keywords: PRRSv, control, elimination, load-close-expose, whole herd exposure, live virus inoculation, serum inoculation, modified-live virus vaccine.

Introduction

Porcine reproductive and respiratory syndrome virus (PRRSv) infection costs from $78 to $255/sow/year to the US swine industry (Holtkamp et al., 2012). Hence, different strategies to control and eliminate PRRSv from infected breeding herds have been implemented. Due to practical reasons and financial advantage over whole herd repopulation, herd closure combined with load-close-expose (LCE) method has become a popular strategy in the North American swine industry to produce negative offspring from PRRSv-positive sow herds (Menard, 2012; Torremorell et al., 2003).

Herd closure consists of interrupting introduction of replacement-gilts for several months (Torremorell et al., 2003) and one alternative is to expose the breeding stock to a replicating PRRSv (Desrosiers and Boutin, 2002; Zimmerman et al., 2012). Options for whole-herd exposure include inoculation of either resident-live virus (LVI) (Batista et al., 2002) or modified-live virus (MLV) vaccine. The general concept of herd closure consists of building protective immunity in the breeding herd, thereby preventing virus transmission to piglets. A recent study reported that the median time from LCE to reach PRRSv stability (TTS) was 26.6 weeks (25th to 75th percentile, 21.6 to 33.0 weeks) (Linhares et al., 2012a).

As part of the stages for PRRSv elimination without depopulation a goal is to produce PRRSv-negative piglets (American Association of Swine Veterinarians - AASV category II-b). Confidence that the herd is PRRS negative is gained if replacement pigs remain PRRSv-antibody free for at least 60 days after introduction into the breeding herd (AASV category III). When the whole herd is PRRSv-antibody free, it

can be considered PRRSv “free” or “naïve” (AASV category IV) (Holtkamp et al., 2011).

There is no information on the success rate of LCE strategy, as indicated by starting with infected breeding herds and achieving category III and subsequently IV. Torremorell documented that herd closure programs without an exposure component had a success rate around 85% for breeding herds endemically infected with PRRSv (Torremorell et al., 2003). Similarly, about 80% of herds that were acutely infected with PRRSv that started LCE program achieved a 90-day period of lack of viremia detection in weaning-age pigs, which is the equivalent to the AASV-classification category II-b (Chapter 2). However, it is still unknown whether herds that reached classification category II-b remained PRRSv-negative after introduction of replacement pigs at the end of the closure period. It has been suggested that management practices influence PRRSv transmission dynamics within infected breeding herds (McCaw, 2000). There is a need to study the effect of specific management practices on success rate for producing PRRSv negative pigs at weaning from infected breeding herds.

The objectives of this study were to: a) determine the proportion of breeding herds that were still infected with PRRSv despite having failed to detect PRRSv in weaned pigs for at least 90 days and, b) to evaluate the association of management practices recommended by veterinarians with success of elimination programs. For the purpose of this study, “successful elimination” was defined as ability of a PRRSv-infected breeding herd that adopted a LCE program to reach AASV category III.

Materials and Methods

Design and source population

This study was the third phase of a prospective study on breeding herds that became infected with PRRSv and adopted herd closure in combination with LCE program using either LVI or MLV vaccine for the whole herd exposure method. There were 56 herds that were followed as part of this study (table 1).

Information for the present study was obtained from a survey sent to veterinarians overseeing the participating breeding herds. The survey collected information regarding PRRSv management and replacement pig flow during and after the closure period (Table 2). The survey also collected data on the frequency that herd veterinarians recommended the management practices indicated by a AASV taskforce as being crucial for herds undergoing PRRSv elimination (Lowe et al., 2012). These practices include the following: change needles between sows and gilts, manure feedback pre-farrow, manure feedback pre-breeding, tissue or serum feedback to gilts, farrowing crates washed and with dry time between litters, change needles and blades between litters, use of warming tubs/split suckle boxes, existence of at least one fall back litter, holding pigs at weaning for quality and litters that were cross-fostered after 24h of birth.

Results obtained during parts 1 and 2 of the study (chapters 2 and 3) were used to better describe the PRRSv infection dynamics on the enrolled herd (Table 3).

Outcomes

This study reported (a) the proportion of herds that achieved TTS that subsequently achieved AASV PRRSv category III status (i.e. “provisional negative” status for PRRSv) as reported by the herd veterinarian, (b) effect of TTS and total production losses on the odds of reaching AASV category III, and (c) proportion of farms that became re-infected with unrelated PRRSv compared to farms that became re-infected with PRRSv that was genetically similar to the previous resident virus. These outcomes were compared between herds that used MLV and herds that used LVI for whole-herd exposure.

Herds that had PRRSv detection 20 weeks after implementation of LCE submitted serum samples for PRRSv RNA amplification using PCR-based methods followed by genetic sequencing of the open reading frame 5 (ORF 5). ORF 5 sequences of PRRSv detected after LCE program were compared to that found prior to the establishment of the program. For the purpose of this study, sequences with nucleotide similarities of 97% or higher were considered “similar”. Viruses that had less than 97% similarities were considered as “new”, or “unrelated” to the resident virus (Murtaugh, 2012).

The association between LCE success and management practices as recommended by the AASV for herds undergoing PRRSv elimination was assessed.

Statistical analysis

To compare frequencies between groups such as percentage of herds that achieved Category III after adopting LCE, or frequency of recommendation of specific management practices cross tabulations with Chi-square or Fisher’s exact test, as appropriate were used. Logistic regression was used to investigate relationships

between herd characteristics (table 2) or management practices with odds of reaching PRRSv AASV category III status. Univariate analysis was conducted for each parameter and p-value of <0.20 was used to screen variables eligible for inclusion in the multivariate model. In the multivariate analysis all possible two-way interactions were assessed and backward selection procedure was used to remove non-significant variables, leaving in the final model variable(s) with p-value of 0.05 or less.

All tests were two-sided, with an alpha level of 0.05 considered to indicate statistical significance. Statistical analyses were performed with the use of SAS 9.2 software (SAS Institute Inc., Cary, NC).

Results

Proportion of herds that achieved AASV category III

Forty herds (71.4 %) that started the LCE program with the goal to eliminate PRRSv reached AASV category III. There was no significant difference between herds that used LVI (70%) compared to those that used MLV (75%) (p = 1.0). There were 3 farms in the LVI group and 3 farms in the MLV group that were infected with an unrelated PRRSv and 9 farms and 1 farm in the LVI and MLV group infected with a virus that was genetically similar to the resident virus detected prior to the adoption of LCE, respectively.

Excluding failures associated with apparent new PRRSv, there still was no significant difference between proportion of farms using LVI (75.7%) and MLV (92.3%) to achieve category III after adopting LCE (p = 0.2584).

From the herds that achieved TTS, 80% and 85% of herds in the LVI and MLV group achieved category III respectively (Fisher's exact $p = 1.0$). Among the ones that did not reach category III, in the LVI group there were 5 farms detected positive with the resident virus and 2 farms detected positive with an unrelated PRRSv. In the MLV group there were 2 farms infected with unrelated PRRSv. Excluding failures associated with detection of new PRRSv, 28 (84.9%) and 12 (100%) farms using LVI and MLV that achieved TTS after adopting LCE subsequently achieved category III, respectively (Fisher's exact, $p = 0.3034$).

Management practices between herds assisted by different veterinary clinics

Herds assisted by different veterinarians or veterinary clinics had different sets of recommendations of management practices to be followed during LCE (Figure 1). "Farrowing crates washed with dry time between litters" was a management practice recommended to all herds. All the investigated management practices were recommended to herds by at least one veterinarian or veterinary clinic.

Association of management practices and ability to reach AASV category III status

The management practice "holding pigs at weaning for quality" was negatively associated with the likelihood of reaching AASV category III. From 7 herds that had the recommendation to follow this practice if needed, 71.4 % failed to reach AASV category III, compared to 22.9 % failures among 37 herds that had the recommendation to not hold pigs at weaning for quality (p 0.0307). The other 9 investigated management practices were not associated with AASV category III status (p-values ≥ 0.4231).

Association of selected herd characteristics and ability to reach AASV category III status

Herds that were infected with RFLP 1-4-4 PRRSv family were less likely to achieve AASV category III status (*p* 0.0285) compared to the rest of the PRRSv isolates in the study. All other herd characteristics were not associated with reaching AASV category III following LCE program (p-values ≥ 0.2011) (Table 4). There was no association between ability to reach AASV category III and a) duration it took to achieve TTS, b) magnitude of production losses from infection to LCE or c) production losses from LCE to recovery of production levels that the herd had prior to PRRSv infection (p-values ≥ 0.3455).

Discussion

This is the first description of the relationship between LCE-associated practices to eliminate PRRSv from breeding herds with the ability of reaching AASV category III for PRRSv status. A better understanding of practices associated with "failures" to eliminate PRRSv from breeding herds will help to increase the success rate of LCE, herein reported to be about 70%.

Torremorell and colleagues reported an 85% success rate for breeding herds that adopted herd closure to eliminate PRRSv. However, those breeding herds were endemically infected with PRRSv and had already reached stability in terms of PRRSv-associated clinical signs at the time when closure was imposed (Torremorell et al., 2003). A previous study by our group showed that herds that adopted an LCE program with LVI with resident virus reached TTS sooner than those that used MLV

(Linhares et al., 2012a). However, the present study reported no significant differences between LVI and MLV herds in the ability to reach AASV category III status, indicating there is no association between duration of time to achieve failure to detect PRRSv RNA by PCR over 90 days and the ability to reach AASV category III status.

This study also described, for the first time, the influence of recommending selected management practices on the ability to achieve PRRSv category III. Knowing which factors are associated with success of PRRSv elimination will allow veterinarians to increase the success rate of PRRSv-elimination through herd-closure methods.

Previous studies by our group reported that herds assisted by a specific veterinary clinic reached PRRSv-stability status sooner and recovered production levels sooner than all other herds. One of the possible explanations is that herds assisted by that particular veterinary clinic could have been implementing key management practices that influenced PRRSv infection dynamics. Therefore, the present study compared recommendation levels of selected management practices between herds of different veterinary clinics and showed that some practices were recommended by all systems whereas some other practices were recommended by some and discouraged by others. Moreover, it was found that herds that held pigs at weaning for quality had lower PRRSv elimination success rate, perhaps because poor doing piglets have higher odds of harboring PRRSv (Cano, 2007).

In some instances lack of farm workers compliance to follow recommendations from veterinarians is possible to occur (Racicot et al., 2012; Racicot et al., 2011).

Therefore there is a need for further studies to evaluate the effect of implementation level (rather than recommendation level) of management practices on ability to successfully eliminate PRRSv from breeding herds using LCE programs.

Monitoring PRRSv in breeding herds undergoing virus elimination by testing 30 preweaning piglets over a 90 day period is a fairly standard practice (Holtkamp et al., 2011). That sampling scheme assumes that PRRSv infection dies out within 90 days once prevalence is below 10%. However, data from this study show that this is not always the case since 5 herds that had previously achieved 90 days of failure to detect PRRSv in pre-weaning piglets failed to reach AASV category III. Therefore, there is the need for further studies to better characterize dynamics of PRRSv infection near the end of elimination programs, documenting duration of PRRSv infection in pig populations at low prevalence, and thus helping veterinarians to plan PRRSv monitoring schemes (sample sizes and frequency of sampling) accordingly.

Evans et al showed that virus persistence at the population level depends on a combination of different herd characteristics. For instance, PRRSv infection persisted longer with increasing herd size, increased contact between different age groups and increased reintroduction of infectious gilts (Evans et al., 2010). In contrast, the present study did not find links between herd demographics and ability to successfully eliminate PRRSv from breeding herds using LCE.

However, our data did show that herds infected with PRRSv belonging to the RFLP pattern 1-4-4 had a significantly lower chance to reach AASV category III than herds infected with PRRSv of other RFLP patterns, suggesting potential existence of

particular PRRSv isolates that have ability to maintain PRRSv infection at low prevalence levels for a relatively longer period of time.

Nevertheless, as reviewed by Murtaugh et al, studies have been conducted to discover genetic markers on PRRSv RNA associated with pathogenicity, but clear predictors of ability of PRRSv to cause disease or to induce pig immune response have not yet been found (Murtaugh et al., 2010). Likewise, there are studies investigating interactions between the pig genome and the PRRSv itself aiming to better elucidate mechanisms that lead to the development of quicker protective immunity against the virus (Abbott, 2012; Murtaugh and Genzow, 2011).

Limitations of this study include that information was collected using a survey and was therefore subjected to recall bias. Moreover, there was no standardized diagnostic monitoring program to be followed by each herd, since each participating herd had influence of their own veterinarian and submitted samples to the diagnostic laboratory of its own choice. Moreover, we investigated the role of recommendation rather than implementation of management practices on ability of herds undergoing PRRSv elimination using LCE to reach AASV category III, so there is the potential for discrepancies between recommendation and implementation levels and ability to successfully eliminate PRRSv.

Conclusions

Although the exposure choice of MLV or LVI can influence production losses and time to achieve TTS, there were no significant differences between these two factors

for achieving PRRSv AASV classification III. Approximately 18% of herds that failed to detect PRRSv in due to wean piglets for 90 days did not achieve AASV category III. More information is needed in herds undergoing PRRSv elimination to decrease this likelihood. In addition, there was no agreement between veterinarians on the relative importance of each management practice and in summary, the variables associated with failure to reach AASV category III were a) being infected with a PRRSv of RFLP pattern 1-4-4 and holding back pigs at weaning for quality.

Acknowledgements

This study was funded by the University of Minnesota's Swine Disease Eradication Center (SDEC).

Tables and figure

Table 1 – Baseline demographic characteristics of herds participating in the study*

Characteristic	LVI	MLV
Number of herds	40	15
Prior immunity	20 (50%)[a]	8 (54%) [a]
RFLP strain 1-4-4	15 (38%) [a]	7 (47%) [a]
Herd size (Mean ± SE)	3,639 ± 298 [a]	2,298 ± 322 [b]
Time from infection to intervention (Mean ± SE)	28 ± 3 [a]	39 ± 9 [a]

* Same letters in the rows indicate no statistical difference between variables among treatment groups (LVI and MLV) at alpha level of 0.05.

Table 2 – Information collected from a survey sent to herd veterinarians participating in the study.

Parameter	**Levels**
Type of PRRSv exposure used in the breeding herd as part of LCE program	Type of immunogen used to expose the breeding herd (LVI or MLV)
Type of PRRSv used in the replacement pigs as part of the LCE program	Type of immunogen used to expose the replacement gilts at the start of the closure period (LVI, MLV or LVI and MLV)
Location of gilts exposure	On-site gilt development unit (GDU), off-site GDU or breeding herd itself
Introduction of gilts during herd closure	Yes or no
Gilt introduction before TTS status was reached	Yes or no
PRRSv exposure status of gilts introduced before TTS status	PRRSv free, exposed to LVI, exposed to MLV, exposed to LVI and MLV
PRRSv exposure status of gilts introduced after TTS status	PRRSv free, exposed to LVI, exposed to MLV, exposed to LVI and MLV
Veterinarian's criteria to decide "end of closure"	Each veterinarian described his or her own criteria
AASV PRRSv status of the breeding herd 60 days after replacement pig introduction (i.e. 60 days after "closure")	Category 1, 2a, 2b, 3 or 4
Similarity of PRRSv ORF-5 sequences at 60 days after closure	Related vs non-related to the PRRSv identified at the beginning of LCE: for herds that had PRRSv detected after LCE, the open reading frame (ORF) 5 was compared with the nucleotide sequence of the PRRS viruses detected prior to start of LCE to the one found after LCE. ORF 5 similarities of at least 97% were considered "related" (Christopher-Hennings et al., 2002; Murtaugh, 2012).
Months negative	Continuous variable (months since herd was considered PRRSv negative since introduction of replacement pigs following LCE program)

Table 3 – Information used from previous studies with the same herds participating in the present study

Parameter	**Description**
Time to PRRSv stability (TTS)	Time from whole herd exposure to either LVI or MLV to achieve 90 days of failure to detect PRRSv in due to wean piglets*
Production losses from infection to LCE	Measured as number of pigs not weaned/1,000 sows according to a statistical process control method using 21 weeks prior to PRRSv detection as reference period**
Production losses from LCE to recovery of production levels that herd had prior to PRRSv infection	Represented by number of pigs not weaned/1,000 sows, according to a statistical process control method using 21 weeks prior to PRRSv detection as reference period**
Previous PRRSv-infection status	'No' for herds that had not been infected with resident or attenuated PRRSv for at least 3 years prior to start of LCE program. 'Yes' for herds with at least one PRRSv infection during that period.
Days from PRRSv-detection to LCE	Number of days from PRRSv diagnosis to date of whole herd exposure to either LVI or MLV vaccine.

* (Chapter 2), ** (Chapter 3).

Table 4 – Characteristics of breeding herds that reached AASV category III for PRRSv status after adopting LCE with goal to eliminate PRRSv.

Parameter	**Number (and proportion of herds that achieved AASV category III after LCE program)**	**p-value**
Type of PRRSv used in the breeding herd as part of the LCE program	LVI: 28 (70%) MLV: 12 (80%)	Fisher's exact test, 0.5209
Type of PRRSv used in the replacement pigs as part of the LCE program	LVI: 24 (67%) MLV: 6 (75%) LVI and MLV: 10 (90%)	Fisher's exact test, 0.3624
Introduction of replacement gilts before the definition of TTS	Yes: 16 (73%) No: 24 (73%)	Chi-square, 1.00
Herds with prior PRRSv infection in the previous 3 years	Yes: 22 (73%) No: 18 (72%)	Chi-square, 0.9120
Replacement pigs were raised on a off site gilt development unit (GDU)	Yes: 12 (80%) No: 28 (70%)	Fisher's exact test, 0.5209
PRRSv detected prior to LCE establishment belonging to the RFLP 1-4-4 family	Yes: 10 (48) No: 30 (88%)	Chi-square, 0.0010**
Period of PRRSv monitoring of piglets during LCE used to define duration of the closure period (i.e. when to introduce replacement pigs in the herd)***	< 90 days: 23 (72%) 90 days: 7 (100%) > 90 days: 10 (63%)	Fisher's exact test, 0.2011

* Number and proportion of herds that reached PRRSv AASV category III after adopting LCE. ** P-value statistically significant at alpha level of 0.05. *** Monitoring PRRSv for a 90-days period is a guideline to achieve AASV category 2b (Holtkamp et al., 2011).

Figure 1 – Profiles of management practices recommendations by herd veterinarians during the herd closure period

	Veterinarians or veterinary clinic recommendation level (%)*															
Managment practices	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Change needles between sows and gilts	100	19	0	50	100	80	50	75	100	25	100	0	100	0	0	0
Manure feedback pre farrow	0	19	0	0	0	100	75	0	0	0	0	0	0	0	0	50
Manure feedback pre breeding	0	19	100	0	0	80	75	25	0	0	0	0	0	0	0	0
Tissue or serum feedback to gilts	0	0	100	0	0	0	75	0	0	0	0	0	0	0	100	0
Farrowing crates washed and with dry time between litters	100	75	100	100	100	40	100	100	100	88	50	100	100	50	100	100
Change needles and blades between litters	100	50	0	50	100	0	100	25	100	13	100	100	100	0	100	100
Use of warming tubs/split suckle boxes	50	0	100	50	0	100	0	0	0	63	100	50	0	0	25	25
Existence of at least one fall back litter	0	38	75	50	25	0	0	0	0	100	0	0	0	0	0	13
Holding pigs at weaning for quality	0	0	0	50	0	0	0	0	0	63	25	0	0	0	0	0
Litters that were X-fostered after 24h of birth	25	6	0	0	0	0	0	0	0	50	0	0	0	25	25	25

* Possible answers were 0, 25, 50, 75 or 100% for each herd. Values represent average between herds assisted by a veterinarian or veterinary clinic.

Chapter V - Effect of modified-live PRRSv vaccine on the shedding of wild-type virus from an infected population of growing pigs

Linhares DCL, Cano JP, Wetzell T, Nerem J, Torremorell M, Dee SA. Effect of modified-live porcine reproductive and respiratory syndrome virus (PRRSv) vaccine on the shedding of wild-type virus from an infected population of growing pigs. Vaccine 2012;30(2):407-413.

Abstract

There are ongoing efforts to eliminate porcine reproductive and respiratory syndrome virus (PRRSv) from regions in the United States swine industry. However, an important challenge for the accomplishment of those efforts is the re-infection of pig units due to the area spread of PRRSv. The objective of this study was to evaluate the effect of PRRS modified-live virus vaccine (MLV) on viral shedding and on dynamics of PRRSv infection in pig populations raised under commercial conditions. The study composed of two rooms of 1000 pigs each. Ten percent of pigs of each room were inoculated with a field isolate of PRRSv. Rooms had separate air spaces and strict scientifically validated biosecurity protocols were adopted to avoid movement of pathogens between rooms. At 8 and 36 dpi (days post inoculation), all pigs of the challenge-vaccine group were inoculated with a MLV vaccine. Pigs of the challenge-control group were placebo-inoculated. Blood and oral fluid samples were collected from each room at 0, 8, 36, 70, 96 and 118 dpi for PRRSv RNA detection using PCR. PRRSv-antibodies were also screened from blood serum samples with a commercially available ELISA test. Additionally, tonsil scraping samples were collected from both groups at 70, 96 and 118 dpi. Moreover, air samples were collected 6 times per week from 0 to 118 dpi and were tested for PRRSv RNA using qPCR assay. There was no difference in the PRRSv infection dynamics measured as duration and magnitude of viremia and seroconversion. Also, there was no difference in the frequency of tonsil scraping samples PRRSv-positive by PCR. However, the challenge-vaccine group had significantly less PRRSv shed compared to the challenge-control group. The challenge-vaccine group had significant less PRRSv-

positive oral fluids at 36 dpi. Moreover, the challenge- vaccine group had significant reduction in the cumulative PRRSv shed in the air.

Keywords: Swine, PRRS, MLV, ATP, vaccine, bioaerosols, shedding.

Introduction

Porcine reproductive and respiratory syndrome (PRRS) is one of the most important swine infectious diseases worldwide. It was first described two decades ago in the United States and Europe (Keffaber, 1989; Wensvoort et al., 1991) and has been reported in many pig producing countries (Zimmerman et al., 2006). The annual losses associated with PRRS in the United States swine industry was estimated at approximately USD$ 664 million (Board, 2011). The etiologic agent, PRRS virus (PRRSv) is a swine specific positive-sense single-stranded enveloped RNA virus belonging to the Arteriviridae family (Benfield et al., 1992).

Upon infection of a susceptible pig, PRRSv replicates extensively in alveolar macrophages (Plagemann and Moennig, 1992), resulting in a viremia that lasts approximately 3 to 5 weeks (Xiao et al., 2004). Nursery pigs are more susceptible than finishing and adult pigs to develop PRRS clinical signs (van der Linden et al., 2003). However, all ages develop viremia and antibody response in the face of PRRSv infection (Klinge et al., 2009; Rossow et al., 1994). Following the cessation of viremia, PRRSv establishes a prolonged infection in lymphoid tissues that has been reported up to 6 months in duration (Wills et al., 1997c). The end of the prolonged infection results in the development of a protective immune response, which clears the infection and confers protection to subsequent challenges (Cano et al., 2007a; Cano et al., 2007b).

PRRSv is shed in different body fluids including oral fluids (Wills et al., 1997b) and bioaerosols (Cho et al., 2006a; Pitkin et al., 2009b). It has been reported that PRRSv remains infectious after being transported for up to 9.1 Km in the air coming out of finishing barns housing pigs infected with PRRSv (Otake et al., 2010). Aerosol transmission is one of the most important PRRSv transmission routes because it is unaffected by standard biosecurity protocols employed by farms such as quarantine, change of coveralls and equipment disinfection. Therefore, farms located nearby sites where PRRSv is prevalent are at risk of PRRSv airborne infection (Desrosiers, 2011; Mortensen et al., 2002). Over the past years, a limited but important number of sow farms in the United States were equipped with air filters to prevent airborne PRRSv infection (Spronk et al., 2010). However, filtering a farm requires substantial investment and growing pig sites are not yet filtered.

To reduce the risk of PRRSv infection of growing pig sites, more information is needed regarding the ability to reduce PRRSv shedding and the resulting viral load into the environment. One potential strategy involves the use of vaccination. Studies have shown that use of modified-live virus (MLV) vaccine prior to PRRSv infection significantly reduces both lung lesions and PRRS clinical signs (Martelli et al., 2009; Okuda et al., 2008). MLV vaccination also reduces PRRSv shedding in semen (Christopher-Hennings et al., 1997) and decreases magnitude and duration of viremia of growing pigs (Zuckermann et al., 2007). Furthermore, Cano et al (Cano et al., 2007a; Cano et al., 2007b) demonstrated that PRRSv MLV vaccine significantly reduced the duration of shedding in animals previously infected with wild-type

PRRSv. However, those experiments were conducted under controlled conditions with a limited number of animals and did not measure aerosol shedding which could represent a risk for neighboring farms.

Therefore, the purpose of this study was to evaluate PRRSv MLV vaccine as a tool to reduce viral shedding as well as duration and magnitude of viremia in infected populations of growing pigs raised under large commercial conditions.

Design and methods

Animals. Two thousand and fourteen piglets of approximately 3 weeks of age were obtained from two PRRSv-free breeding herds of the same production system consisting each of 6,000 C-42 sows (Pig Improvement Company) crossed with PIC 410 boar line (Pig Improvement Company). The source breeding herds were vaccinated with Inglevac® CircoFLEX (Boehringer Ingelheim Vetmedica Inc, St. Joseph, MO), Ingelvac® MycoFLEX (Boehringer Ingelheim Vetmedica Inc) and Flusure XP (Pfizer Animal Health, St. Joseph, MO). All piglets received at weaning a full dose of Ingelvac® CircoFLEX and Ingelvac® MycoFLEX vaccines. All the procedures with animals described in this study were previously approved by the University of Minnesota Institutional Animal Care and Use Committee (IACUC).

Housing. Pigs were housed in the Pipestone Veterinary Clinic's tunnel ventilated research barn located in Pipestone, Minnesota. The research barn consisted of two rooms (west and east) separated by a solid wall and each room had independent attic and slurry pit. Both rooms had 42 pens of 2.50 x 7.0m (17.5m^2), which were divided

by metal gates. There were 23 to 25 pigs per pen. Waterers and feeders of both rooms were standardized across pens and rooms. The north side of each room was equipped with two 91 cm and four 122 cm of diameter wall exhaust fans and two 50.8 cm and 61 cm diameter pit fans. Figure 1 illustrates the location of rooms in the barn as well as the entry points and place where air samplings took place.

Pig care. All piglets were injected 0.1 mL of Excede (Pfizer Animal Health) at birth. One week after weaning, the following therapy was provided to pigs via water: 32 oz/gal of vitamin E 40% (400 IU of Vitamin E per gram, AHC Products, Inc), ileitis vaccine (Boehringer Ingelheim Vetmedica Inc) and erysipela vaccine (Boehringer Ingelheim Vetmedica Inc). The day before PRRSv experimental infection (-1 dpi), pigs were administered via water 4 oz/gal of ProPrin LQ (12% acetylsalicylic acid, AniMed AHC Products, Inc) and 0.67 pkg/gal of PenAqua Sol-G (0.5 billion I.U. Penicillin G Potassium, Bimedia, Inc).

PRRS virus isolate. A PRRSv of restriction fragment length polymorphism (RFLP) pattern 1-18-2 belonging to Dr Fred Leung's Lineage 1, sublineage 5 group (Shi et al., 2010b), which is the same lineage as MN 184 type viruses, was isolated from a pig production system located in South Dakota experiencing PRRSv-associated clinical signs. In that system, abortions were at levels below 2.0% and pre-weaning mortality was below 10% for 12 months before the outbreak and the pigs were considered PRRSv-free based on ELISA from sows and RT-PCR from piglets. During winter 2009 sows became off-feed, abortions spiked to 30% and pre-weaning mortality reached 40%. Piglets became PRRSv-RNA positive by PCR and sows seroconverted

to ELISA (IDEXX PRRS 2XR Ab Test, Westbrook, ME). A serum sample was drawn from a 20 days-old male piglet with poor hair coat and submitted to the University of Minnesota's Veterinary Diagnostic Laboratory (MN-VDL). After 2 passages in a porcine alveolar macrophage (PAM) cell line, the isolate had a viral concentration of 10^7 $TCID_{50}$/ml. The inoculum was screened by PCR-based assays at the MN-VDL and tested negative for adventitious agents including porcine circovirus type 1, porcine circovirus type 2, porcine parvovirus, bovine viral diarrhea virus and pseudorabies virus.

Experimental design. Upon arrival, pigs were sorted by gender, source and size and were randomly allocated to the west and east rooms. One thousand and five pigs were housed in the west (hereafter called challenge-control room) room and 1,009 were housed in the east (challenge-vaccine room) room. At 8 weeks of age, 100 pigs of each room were conveniently selected (2 - 3 pigs/pen) and intramuscularly inoculated with 2 mls of PRRSv inoculum. At 8 and 36 days post infection, all pigs of the challenge-vaccine group were vaccinated with 2ml of Ingelvac® PRRS ATP (Boehringer Ingelheim Vetmedica Inc, St. Joseph, MO) vaccine. At the same time, pigs of the control room were sham-inoculated with 2ml of saline solution.

Environmental conditions. The room temperature in the challenge-control group averaged 23.0ºC from 0 to 118 days post experimental inoculation (DPI) (range 18.5 - 27.2). In the challenge vaccine group, the room temperature averaged 23.1ºC (range 19.1 – 27.5) in the same period.

Monitoring of PRRSv infection

Animal sampling. The protocol used to monitor PRRSv infection is illustrated in table 1. Briefly, 35 piglets of each group were randomly selected (no more than 1 pig per pen), identified with ear-tag and designated as “monitor pigs”. Blood samples were drawn from monitor pigs into sterile blood vacutainer tubes (Becton-Dickinson Vacutainer, Franklin Lakes, NJ) via jugular venipuncture at -7, 0, 8, 36, 70 96 and 118 days post infection (DPI) and submitted for PRRSv RT-PCR (Wasilk et al., 2004) in pools of 5 and PRRSv ELISA tests individually (Herd Chek PRRS Antibody 2XR Test Kit, IDEXX Laboratories, Westbrook, ME). Additionally, 12 pens of each room were randomly identified and had oral fluids collected as described by Prickett et al. (Prickett et al., 2008) at the same points in time and submitted for PRRSv RNA detection by RT-PCR (Kittawornrat et al., 2010). Also, 12 monitor pigs were randomly selected in each room and had their tonsils scraped at days 70, 96 and 118 DPI (Table 1) using the technique described by Wills et al. (Wills et al., 1997a) and submitted for PRRSv RNA detection using RT-PCR (Kittawornrat et al., 2010).

Air sampling. Two identical liquid cyclonic collectors (Midwest MicroTek, Brookings, SD) capable of collecting 400 liters of air per minute were used as described by Dee et al. (Dee et al., 2009) to collect air coming out of each room (challenge-control and challenge-vaccine) every day from 08:00 AM to 08:30 AM from 0 to 118 days post infection except Sundays. The cyclonic collectors were

placed at 1.6 meters above the ground, being at the distance of 2 meters from and at the same high of the center of the 122 cm of diameter exhaust fan. One cyclonic collector was used for each room. The air samples were kept frozen at -20ºC for up to 10 days and then frozen at -80ºC until tested for PRRSv RNA detection by RT-PCR (Dee et al., 2009).

Sample testing. The PRRSv RT-PCR and PRRSv ELISA tests from serum samples were performed at the Boehringer Ingelheim Vetmedica Inc Health Management Center, Ames, Iowa. The RT-PCR for detection of PRRSv RNA from oral fluids and tonsil scrapping samples were performed at the Iowa State University Veterinary Diagnostic Laboratory (ISU VDL), Ames, Iowa. The PRRSv quantitative RT-PCR from air samples were performed at the University of Minnesota Veterinary Diagnostic Laboratory (UMN VDL), Saint Paul, Minnesota. All serum, oral fluids and tonsil scraping samples that were PRRSv RNA positive in the RT-PCR test were submitted for open reading frame 5 (ORF 5) nucleotide sequencing at the ISU VDL. The air samples that were PRRSv RNA positive by RT-PCR were submitted to the UMN VDL for ORF5 nucleotide sequencing.

Biosecurity procedures. Extensive effort was placed on biosecurity to reduce the risk of transferring the vaccine virus from one room to another using scientifically validated protocols of biosecurity (Dee et al., 2004b; Pitkin et al., 2009a). The inlets were equipped with insect screening. Each room of the barn had a designated entrance involving a shower-in shower-out facility and room-specific coveralls, boots and equipment. For air collection, there was a dedicated cyclonic collector for each

room, which was disinfected everyday before and after use according to protocols previously validated (Dee et al., 2009).

Growing performance. Frequency of mortality, removal rate and antibiotic treatment were compared between groups. Moreover, feed efficiency and average daily gain were computed from 37 to 137 days of life and compared between groups.

Statistical analysis.

The Fisher's exact test was used for between-groups comparison of the difference on the frequency of positive results for serum ELISA, serum PCR, oral fluids PCR and tonsil scrapings PCR. To compare the average of ELISA S/P ratios, the paired t-test was used. To compare the frequency of air positive days the McNemar's test was used, and to compare the RNA copies/ml concentration of air positive days and the distribution curve of positive days the Mann-Whitney-Wilcoxon test was used. For the growing performance comparisons, frequency of mortality, removal to sick pens and injectable antibiotic use were compared using Chi-squared test and average daily gain and feed efficiency were compared with paired t-test. All the analyses were performed using SAS 9.2 statistical software and the level of statistical significance considered was of $\alpha = 0.05$.

Results

Establishment of infection: The population of pigs used in this study was confirmed to be serum- and oral fluids-PRRSv RNA negative by RT-PCR assay as well as anti-

PRRSv antibody negative at 5 days before challenge. The experimental infection was successfully established as determined by serocoversion of 1/3 of monitor pigs at 8 dpi, detection PRRSv viremia in all monitor pigs at the same point in time and 100% seroconversion of monitor pigs at 36 dpi.

Effect of the vaccine on PRRSv infection dynamics: There was no significant difference between groups in the PRRSv infection dynamics measured as viremia level and duration, antibody response and presence of PRRSv RNA in the tonsils (Table 2).

Effect of the vaccine on viral shedding via oral fluids: PRRSv RNA was detected in oral fluids of both rooms at 8 dpi. At 36 dpi, the proportion of PCR-positive oral fluid samples was significantly lower in the challenge-vaccine group (p= 0.03). After 36 dpi, all collected oral fluids samples from both groups were PCR-negative.

Effect of the vaccine on viral shedding in bio-aerosols: The summary of PRRSv RNA findings in air samples by qRT-PCR are summarized in table 3 and illustrated in figures 2-4. There was no difference in PRRSv RNA concentration in the air samples between groups. However, the challenge-vaccine group had a significant reduction in the total number of days that PRRSv RNA was detected in air samples (14 days, p = 0.0004). Moreover, the challenge-vaccine group had a shorter period of time (25 days) that had air samples positive for PRRSv RNA.

PRRSv differentiation: To determine whether the PRRSv RNA detected in each sample was from the (a) PRRSv isolate used for experimental challenge, (b) from the

vaccine virus or (c) from new lateral virus introduction, PRRSv PCR-positive samples were submitted for virus differentiation by ORF 5 sequencing. The PRRSv ORF 5 sequence phylogenetic comparison of the sequenced samples is shown in figure 5.

In both groups, the PRRSv RNA detected in the air, serum and tonsil scraping samples were at least 99.0% similar to the wild type virus based on ORF 5 nucleotide sequences, with the exception of the air samples detected in the challenge-vaccine group at 15 and 16 DPI (7 and 8 days post first vaccination) that were identical to vaccine strain (Figure 5). The viruses detected in oral fluids could not be differentiated.

Effect of the vaccine on growing performance: There was no difference in the number of deaths (*p* 0.58), number of pigs removed to sick pens (*p* 0.33) and number of pigs treated with injectable antibiotics (*p* 0.98) between groups. The overall mortality in both groups was 1.5%. The average daily gain (from 37 to 113 days) in the challenge-control and challenge-vaccine group was 1.70 and 1.68 respectively (*p* 0.42), and the feed conversion rate for the same time period was respectively 2.18 and 2.20 (*p* 0.22). In summary, despite the evidence of PRRSv infection, there was no apparent PRRS clinical manifestation in the study groups.

Discussion

Validation of strategies to minimize PRRSv shedding are essential to prevent PRRSv dissemination and to assist in the regional control and elimination of PRRSv. This

study evaluated, for the first time, the effectiveness of a PRRS modified-live virus vaccine to reduce viral shedding through oral fluids and bio-aerosols in an infected large growing pig population under field conditions. The study was conducted in a research wean-to-finish barn that resembles typical commercial growing pig farms located in the Mid Western United States.

The PRRSv inoculum used in this study was isolated from a pig production system that attributed extreme piglet mortality and reproduction problems to the strain used (RFLP pattern 1-18-2). Moreover, the veterinarian responsible for that production system reported that this same strain (with similar ORF 5 sequence) was detected in multiple unrelated farms suggesting its potential airborne transmission capacity. PRRSv ORF 5 sequencing (Figure 5) showed 100% nucleotide similarity between the production system, the inoculum that was prepared for this study and viremic pigs at 2 and 8 days post experimental infection, which suggested that the inoculum used in this study was similar to the one that caused infection and economic impact in the aforementioned pig production system.

To mimic PRRSv infection in commercial pig barns, 10% of piglets were inoculated with a live PRRSv isolate. At least two pigs per pen were infected to assure that all non-infected pigs had a chance to make effective contacts with inoculated pigs and become infected naturally. Regarding the PRRSv infection dynamics observed in the study, the timing of viremia was in line with the expected based on previous reports (Cano et al., 2007a; Park et al., 1995), and seroconversion happened sooner than reports from other studies (Nelson et al., 1994; Park et al., 1995; Yoon et al., 1995).

Considering that PRRSv causes prolonged infection in lymphoid tissues (Wills et al., 1997c), in this study tonsil scrapings were collected from monitor pigs at 70, 96 and 118 dpi to have an assessment of the effect of Ingelvac® PRRS ATP vaccine on the duration of PRRSv infection. There was no difference in the number of PRRSv RNA-positive tonsil scraping samples between groups.

Although the frequency of oral fluid samplings was limited, under the conditions of this study the oral fluids PRRSv RT-PCR results appeared to be correlated with detection of PRRSv qPCR results in air samples. At around 36 dpi the control group had 6 out of 6 oral fluids PCR positive and the air samples were frequently positive around 36 dpi. One month later (at 70 dpi) all oral fluids were negative. After 70 dpi there were no days with air samples PCR-positive. Likewise, in the challenge-vaccine group at 36 dpi 2 out of 6 oral fluid samples were positive and the last PCR-positive air sample in the challenge-vaccine group was soon after that, at 45 dpi. Further studies with higher frequency of oral fluid samplings that also include air samplings are necessary to further investigate this association.

Results showed that although vaccine virus established infection, the wild-type isolate remained the dominant isolate in the challenge-vaccine group population. In other words, the vaccine isolate could not replace wild-type virus under the conditions of this study, which has also been reported by others (Cano et al., 2007a; Christopher-Hennings et al., 1997). Taking all this information together, we speculate that when PRRSv-positive pig populations are vaccinated with PRRS modified-live virus

vaccine, it is expected that the wild-type virus remains dominant and that there is limited regional spread of vaccine virus through bio-aerosols.

The PRRSv ORF-5 sequence phylogenetic comparison of sequences (Figure 5) illustrates that in this study the PRRSv changed less than 2% across the 118 days post infection, which was the rate of change that was expected. The estimated rate of nucleotide substitution for PRRSv is from 4.7 to 9.8 x 10^{-2} per year (Hanada et al., 2005; Jenkins et al., 2002), the highest ever reported among all RNA viruses. We did not investigate the reason why oral fluids samples could not be differentiated.

Under the conditions of this study, vaccinating with PRRS MLV a population of previously infected pigs did not negatively impact growing performance (no difference in mortality, average daily gain and feed efficiency between groups). This finding supports the practice of administering PRRSv MLV vaccine preventively in growing pigs housed in sites with high risk of PRRSv-infection (i.e. high density regions).

Therapeutic vaccination, as described in this study, is another option that can be used to reduce PRRSv regional dissemination from infected pig populations. A possible explanation for the effect of vaccination on viral shedding reduction is that even though PRRSv is highly infectious, it is not highly contagious which means that once PRRSv infects a growing pig population, it takes time to spread in the barn. Conversely, when a live virus is inoculated in all pigs, at once every animal is removed from the “susceptible” pool and is given the chance to develop immunity. In

other words, at the population level, the duration of shedding is expected to be less in populations mass-exposed with a live virus because the probability of transmission is reduced as a result of immunity (Cano et al., 2007a).

It is reasonable to speculate that the effect of one dose of attenuated vaccine would be similar to that of two doses. The benefit of two shots is to assure that the right dose was effectively administered in 100% of the pigs. Future studies are needed to further evaluate that.

Regarding type of vaccine, up to this moment, there is no scientific evidence that killed PRRSv vaccine can develop protective immunity or reduce shedding in pigs (Murtaugh and Genzow, 2011). Ingelvac® PRRS ATP vaccine was chosen as attenuated vaccine for this study because it was the product being used by the production system that participated in this study. Considering that the degree of genetic homology of PRRSv ORF5 between challenge isolate and attenuated vaccine is not a good predictor of vaccine efficacy (Prieto et al., 2008), we speculate that similar results could be expected if Ingelvac® PRRS MLV (11.3% divergent from Ingelvac® PRRS ATP based on ORF5-sequence) or another attenuated PRRSv vaccine was used.

The strengths of this study included that the experiment was performed under field conditions in a population of 2,000 pigs with a PRRSv isolate that was circulating and causing clinical signs associated with PRRS in that region. This study shows for the first time that PRRSv modified-live virus vaccine is an effective tool to reduce the

PRRSv load in the air produced by PRRSv infected pig populations. Pigs from the vaccinated group had reduced frequency of air PRRSv positive days (p 0.0004).

This study had some limitations, which include the fact that there was a single air sampling per day that collected about 12,000 liters of air coming out of each room (group). Likewise, oral fluids and serum sampling took place only on a monthly basis, which was enough to answer the study's main question but did not allow proper correlation between the outcomes of the different techniques (PCR from serum, oral fluids and air). Multiple air samplings per day and increased frequency of blood and oral fluids collection would give more statistical power and be more representative of the true PRRSv infection and shedding dynamics. Moreover, although there were more than 1,000 pigs per group, the study as a whole was a single replicate that evaluated the effect of the treatment (vaccine) against a single PRRSv wild-type isolate and with a single pig genetic line. Therefore, futures studies to address those limitations would be of great value.

Conclusions

This study reports for the first time the use of PRRSv modified-live virus vaccine as a tool to reduce viral shedding to the environment, measured with PRRSv RT-PCR from oral fluids and bio-aerosols collected with a previously validated air sampler device. Despite the limitations of the study, results support that vaccinating infected growing pig populations with PRRSv MLV vaccine is a potential method to enhance the success of PRRSv regional control because it significantly reduces the cumulative

virus shed in the air. Additional studies using different PRRSv strains and different field conditions would be helpful to further understand the findings reported in this study.

Acknowledgements

This study received funds from University of Minnesota Swine Disease Eradication Center (SDEC), from Boehringer Ingelheim Vetmedica Inc and from Pipestone Veterinary Clinic.

Tables and figures

Table 1 – Pig sample collection and testing time scheme*.

	DAYS POST INOCULATION						
	-7	0	8	36	70	96	118
Serum PCR	X	X	X	X	X	X	X
Serum ELISA	X	X	X	X	X	X	X
Oral fluids PCR	X	X	X	X	X	X	X
Tonsil scraping					X	X	X

* In addition, air samples were collected out of exhaust fans of both groups 6 days a week from day 1 to 118 post experimental inoculation (dpi).

Table 2 – PRRSv infection monitoring qualitative results at different points in time shown as number of positive results / total number of tested samples (percentage of positive results).

Sample	Group¶	DAYS POST INOCULATION						
		-7	0	8	36	70	96	118
Serum PCR§ Positive/total (% positive)	Ctrl	0/7 (0.0%)	0/7 (0.0%)	7/7 (100%)	4/7 (57%)	0/7 (0.0%)	0/7 (0.0%)	0/7 (0.0%)
	Vac	0/7 (0.0%)	0/7 (0.0%)	7/7 (100%)	4/7 (57%)	0/7 (0.0%)	0/7 (0.0%)	0/7 (0.0%)
Serum ELISA Positive/total (% positive)	Ctr	0/35 (0%)	0/35 (0%)	11/35 (31%)	35/35 (100%)	35/35 (100%)	33/34 (97%)	32/34 (94%)
	Vac	0/35 (0%)	0/35 (0%)	12/35 (34%)	35/35 (100%)	35/35 (100%)	34/34 (100%)	34/34 (100%)
Oral fluids PCR Positive/total (% positive)	Ctr	0/7 (0.0%)	0/7 (0.0%)	2/6 (33%)	6/6* (100%)	0/7 (0.0%)	0/7 (0.0%)	0/7 (0.0%)
	Vac	0/7 (0.0%)	0/7 (0.0%)	4/6 (66%)	2/6 (33%)	0/7 (0.0%)	0/7 (0.0%)	0/7 (0.0%)
Tonsil scrapping PCR Positive/total (% positive)	Ctrl	-	-	-	-	6/12 (50%)	5/12 (42%)	2/12 (17%)
	Vac	-	-	-	-	4/12 (33%)	5/12 (42%)	1/12 (8%)

§ samples tested for PRRSv RNA by PCR in pools of 5.

¶ Ctrl = Challenge-control group; Vac = Challenge-vaccine group.

* Superscript mark indicates statistical significance between groups at alpha level of 0.05.

Table 3 – Summary of PRRSv RNA detected by qRT-PCR from air samples.

	Challenge-control group	**Challenge-vaccine group**
PRRSv concentration (RNA copies/mL)		
mean	4.3×10^8	4.5×10^8
minimum	1.2×10^4	4.9×10^3
maximum	1.9×10^9	2×10^9
95% conf interval	(2.6×10^8, 6.1×10^8)	(2.7×10^8, 8.1×10^8)
Frequency of air-PRRSv qPCR positive days		
Number of positive days	31	17*
Duration (last positive day)	70	45

* Superscript mark indicates statistical significance between groups at alpha level of 0.05.

Figure 1 – Illustration of the barn, rooms, entry rooms and air flow. Moreover, the "X" show points where air collectors were placed.

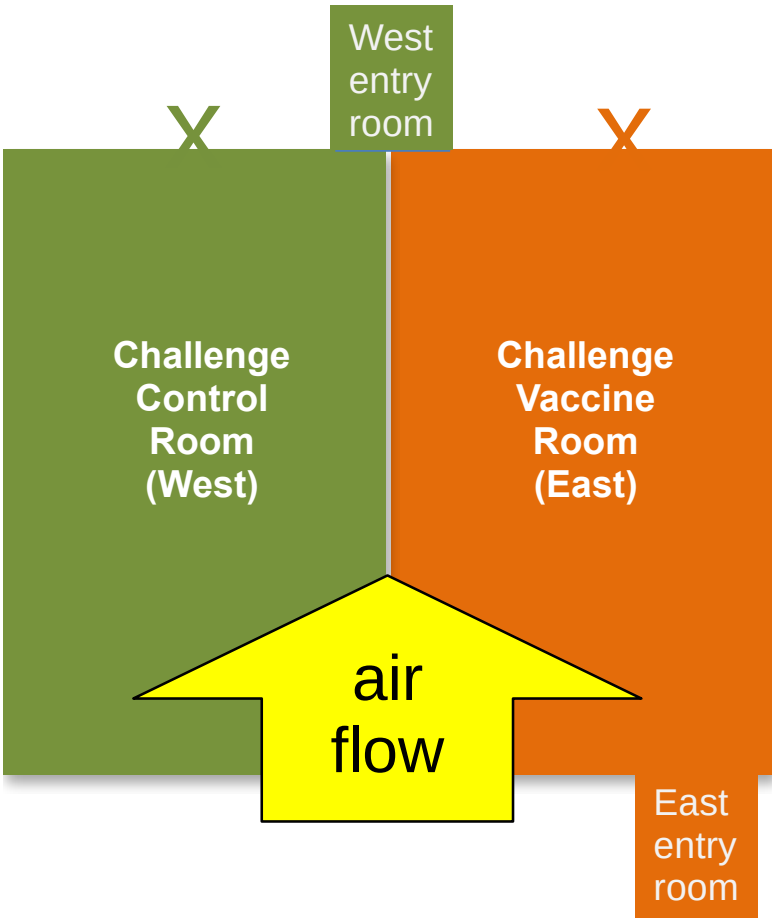
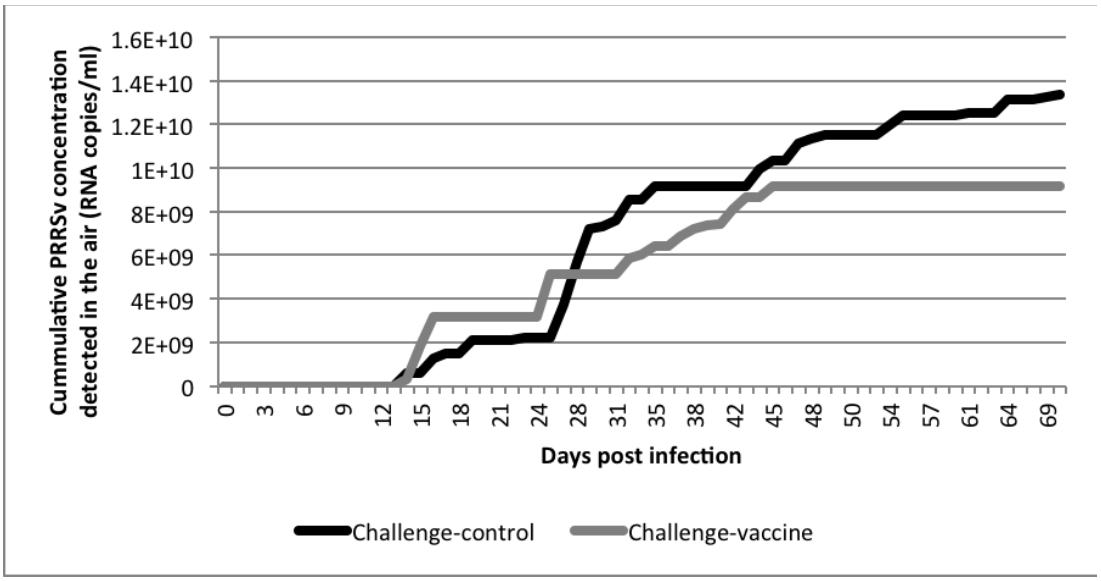


Figure 2 – Cumulative days of PRRSv RNA* detected in air samples by RT-PCR.



Calculated as cummulative sum of PRRSv RNA copies/ml detected in air samples from each group over time.

Figure 3 - Probability curves of control and treatment group to have air positive days

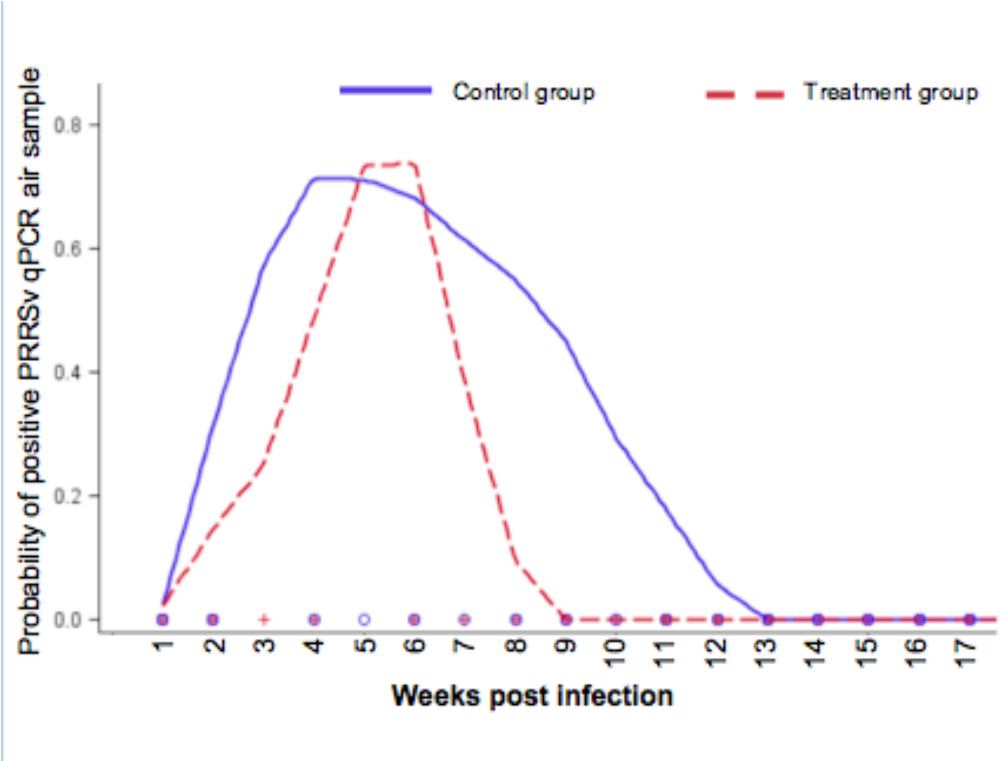
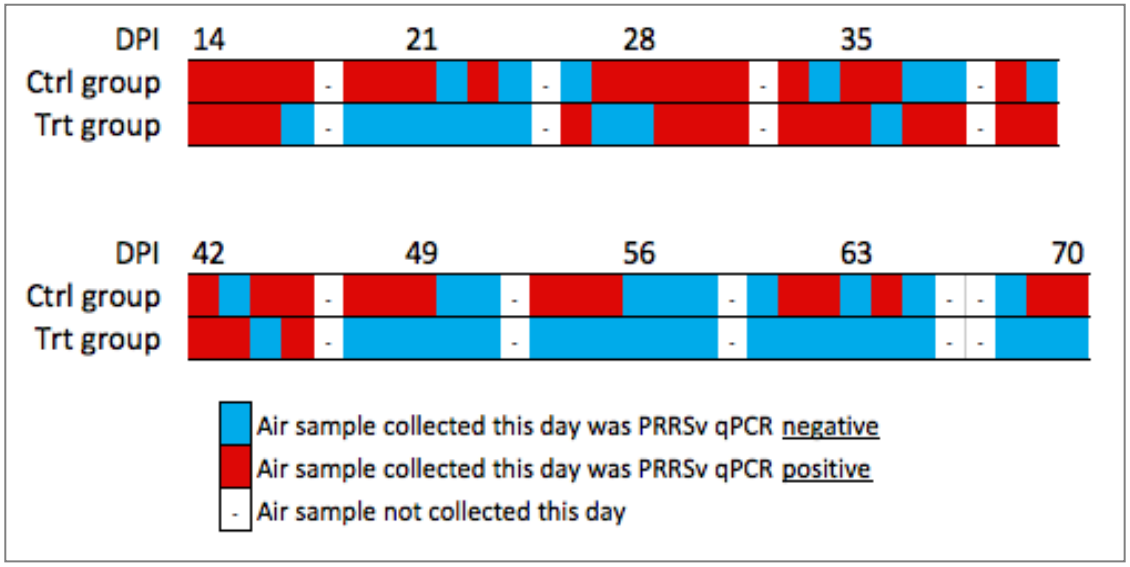
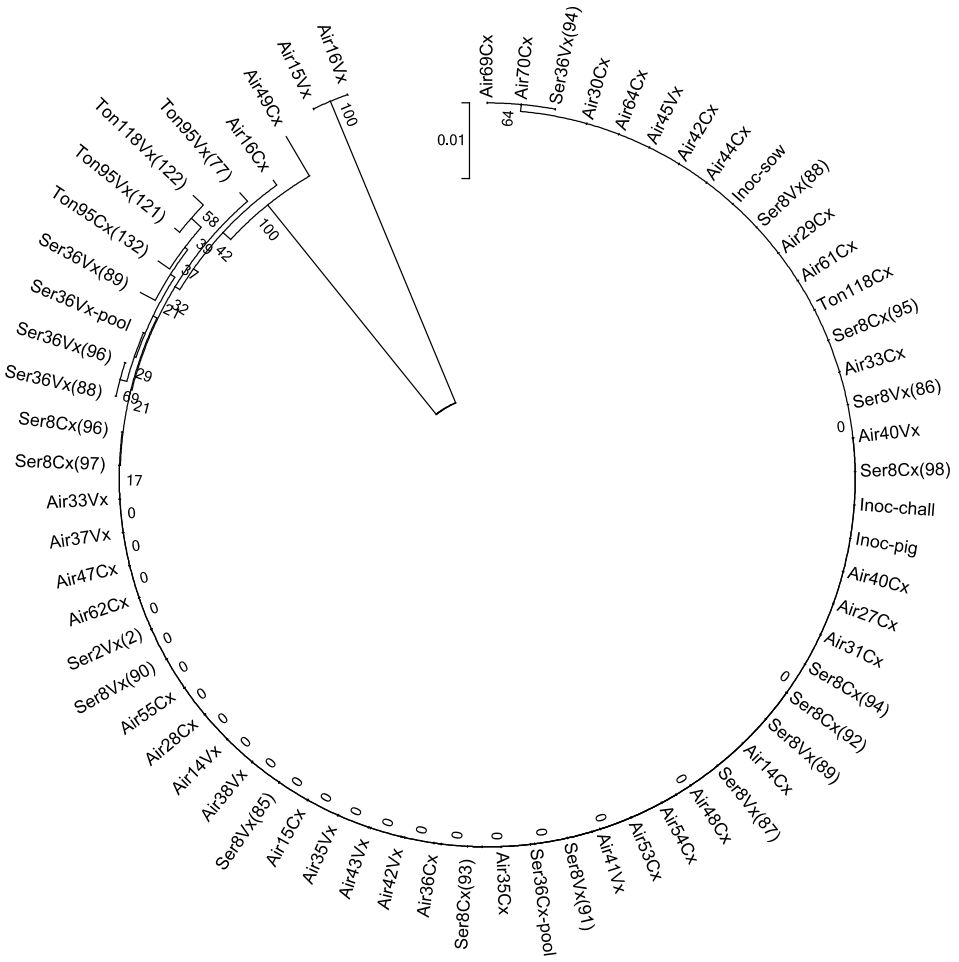


Figure 4 – Detailed air sample PRRSv qPCR results from 14 to 70 days post experimental infection*.



* All samples collected from 0 to 13 and after 70 dpi resulted PRRSv RNA-negative by qRT-PCR.

Figure 5 – Phylogenetic comparison between PRRSv-ORF 5 nucleotide sequences*



* The evolutionary history was inferred using the Neighbor-Joining method [37]. The optimal tree with the sum of branch length = 0.10219240 is shown. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Maximum Composite Likelihood method [38] and are in the units of the number of base substitutions per site. The analysis involved 66 nucleotide sequences. All positions containing gaps and missing data were eliminated. There were a total of 591 positions in the final dataset. Evolutionary analyses were conducted in MEGA5 [39].

Chapter VI - General discussion and conclusions

Controlling PRRSv dissemination from infected populations is crucial to reduce virus spread within a region and thereby facilitate PRRSv regional elimination programs.

This thesis showed that immune management strategies can be employed in breeding herds and growing pig populations to minimize PRRSv dissemination and PRRSv economic impact.

The introduction and first chapter of this thesis provide background information on PRRSv and discuss the importance and need to better understand which methods are available to control and eliminate PRRSv from pig populations. The subsequent chapters evaluated immune management strategies to eliminate PRRSv from breeding herds, assessed the impact of those strategies in production losses, evaluated management risk factors associated with successful PRRSv elimination or failure, and lastly evaluated vaccination as means to reduce virus dissemination from infected growing pig populations.

Chapter 2 compared the use of two immunogens as part of load-close-expose programs in regards to the time it takes to reach PRRSv stability (TTS) from infected breeding herds. Results showed that despite the great variability of TTS between enrolled herds, there were some variables associated with shorter TTS. Specifically, herds that used LVI as the method of whole-herd exposure reached TTS earlier than herds that used MLV vaccines. Moreover, herds with history of PRRSv infection in the 3 years prior to our study reached TTS sooner than herds without history of PRRSv in those same 3 years. Furthermore, herds assisted by a specific veterinary clinic reached TTS sooner than all other herds suggesting that there might be

management practices associated with shorter TTS. Our results also showed that PRRSv-monitoring must be done repeatedly over time to increase confidence of PRRSv-negative status of weaned piglets. Altogether, those findings represent prove of concept that herds can reach PRRSv stability as soon as 84 days from establishment of LCE. Further studies are needed to better understand specific factors associated with reaching short TTS, such as farm layout, pig flow and implementation of specific management practices during the period of herd closure.

Chapter 3 evaluated the effect of attenuated PRRSv vaccine inoculation compared to the use of live-virulent virus inoculation on production performance in breeding herds. It was shown that herds that used MLV vaccine as part of load-close-expose herd closure programs recovered production levels and had a less severe production impact than herds that used LVI. Also, herds that reported previous PRRSv infection reached time to baseline production (TTBP) sooner than herds with no history of PRRSv infection in the previous 3 years. Interestingly, herds assisted by a specific veterinary clinic recovered production faster than other herds, raising the hypothesis that specific management practices could be associated with herd closure effectiveness.

The chapter 3 addendum consisted of economic models to assist veterinarians to make informed decisions between LVI and MLV as part of LCE program to eliminate PRRSv. Taking in consideration the future prices for market hogs and pig feed of the next 12 months, and $13.52 loss per pig that is PRRSv-positive, the results suggested that MLV would be a better economical choice.

Chapter 4 was a follow-up from the previous 2 chapters and investigated herd-closure practices associated with successful PRRSv-elimination from breeding herds. It showed that the success rate of LCE programs was 76% and 92% for LVI and MLV herds, when failures associated with unrelated PRRSv were excluded from the analysis although these differences were not statistically significant. Moreover, one out of six herds that achieved 90 days of failure to detect PRRSv in due to wean piglets failed to achieve AASV category III, which indicates that more strict monitoring programs should be adopted in herds undergoing PRRSv elimination. Interestingly, herds assisted by different veterinary clinics had different set of recommendations of management practices to be followed during LCE, indicating that there was no agreement between veterinarians on what is the relative importance of each management practice evaluated. The variables associated with failure to reach AASV category III were a) being infected with a PRRSv of RFLP pattern 1-4-4 and b) holding back pigs at weaning for quality.

Chapter 5 focused on PRRSv-infected growing pig populations. Growing pig populations are a major source of PRRSv dissemination within regions. The effect of an attenuated PRRSv vaccine was evaluated on shedding and generation of aerosols of a wild-type virus from an infected growing pig population raised under Minnesota's swine industry field conditions. Our results showed that vaccinating acutely PRRSv infected growing pigs with MLV vaccine reduced duration and magnitude of PRRSv detection in aerosols exhausted from the pig barn.

Further studies are needed to characterize the effect of implementation (rather than recommendation) of key management practices on the success of PRRSv elimination programs. There is also a need to better understand the procedures to prepare live (virulent) virus inoculum to expose adult animals in load-close-expose programs. For instance, there is a need to document the effect of PRRSv infectivity dose, route of exposure and materials used to dilute the serum on TTS and TTBP. On the growing pig side, there is a need to validate other effective methods to decrease virus dissemination from infected pig sites that continue to pose a risk for neighboring pig populations, for example by inactivating PRRSv in aerosols coming out of infected pig barns.

Altogether, the work presented in this thesis provided science-based information on immune management strategies to control and eliminate PRRSv infection from pig populations. Information herein reviewed, discovered, described and discussed has direct implications to the swine industry and can assist veterinarians to choose appropriate strategies to reduce production losses and decrease PRRSv dissemination from infected pig populations.

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